VAJIRAM & RAVI

Institute for IAS Examination

PowerUp

Current Affairs Test Series – 2025

CA Test – 12 – (CA2262)

Answer Key &
Detailed Answer Explanations

CA Test-12 (CA2262) - Answer Key									
1. (b)	11. (a)	21. (c)	31. (a)	41. (c)	51. (d)	61. (c)	71. (c)	81. (b)	91. (a)
2. (a)	12. (c)	22. (c)	32. (d)	42. (c)	52. (a)	62. (c)	72. (d)	82. (a)	92. (c)
3. (d)	13. (b)	23. (b)	33. (b)	43. (d)	53. (d)	63. (b)	73. (b)	83. (b)	93. (c)
4. (a)	14. (d)	24. (d)	34. (b)	44. (c)	54. (a)	64. (d)	74. (a)	84. (c)	94. (b)
5. (d)	15. (d)	25. (b)	35. (d)	45. (b)	55. (d)	65. (d)	75. (c)	85. (a)	95. (d)
6. (d)	16. (a)	26. (c)	36. (c)	46. (d)	56. (c)	66. (d)	76. (c)	86. (c)	96. (a)
7. (a)	17. (b)	27. (b)	37. (b)	47. (b)	57. (c)	67. (a)	77. (d)	87. (d)	97. (a)
8. (b)	18. (b)	28. (a)	38. (c)	48. (c)	58. (a)	68. (c)	78. (c)	88. (a)	98. (b)
9. (d)	19. (c)	29. (c)	39. (d)	49. (c)	59. (c)	69. (c)	79. (c)	89. (d)	99. (d)
10. (a)	20. (b)	30. (b)	40. (c)	50. (a)	60. (a)	70. (a)	80. (a)	90. (d)	100. (b)

Q1.
Answer: b
Explanation:

- **A metal** is a material defined by specific properties, including conducting electricity well. Every metal has a finite conductivity. It changes when the metal is heated or cooled. Scientists know of many metals that have a finite conductivity at room temperature and infinite conductivity at very low temperatures.
- Some metals, or metallic substances, do something weird. Below the critical temperature, their electrons experience the net attractive force and pair up — but then **they don't yet condense to form a superconductor**. That is, the material doesn't become a superconductor but **just a better conductor, and it conducts electricity with Cooper pairs, not electrons.** In this state, the material is said to have **become a Bose metal.**
- A **Bose metal** is a kind of anomalous metallic state (AMS). **The 'anomaly' is that the Cooper pairs are formed but don't condense into a superconductor.** In technical terms, they fail to establish long-range superconducting coherence.
    - Cooper pairs are a pair of electrons with opposite spins that are loosely bound at absolute temperatures due to electron-lattice interactions. Superconductivity occurs when electrons spin into so-called Cooper pairs, which can travel through the lattice together.
- Traditional theories that describe disordered metals say that at absolute zero temperature, the metals should have either zero conductivity (become an insulator) or infinite conductivity (become a superconductor). A Bose metal challenges this description because **its conductivity is between zero and infinity as the temperature tends to absolute zero. So, option b is correct.**

Therefore, option (b) is the correct answer.
Relevance: A team of researchers from China and Japan reported in the journal Physical Review Letters that they had found strong signs that niobium diselenide (NbSe2) can become a Bose metal.

Q2.
Answer: a
Explanation:

- The **Department of Biotechnology (DBT)** established Biotechnology Industry Research Assistance Council (**BIRAC**) in 2012 as an interface agency to nurture and strengthen the start-up ecosystem in the country with a focus on the biotech sector.
- BIRAC supports biotech start-up ecosystem by establishing bio-incubators as well as through various funding schemes such as:
    - **Biotechnology Ignition Grant (BIG) provides financial support of up to INR 50 lakhs for a period of 18 months to early-stage start-ups** and individual entrepreneurs to establish Proof of Concept (PoC) for their innovative ideas. Close to 1000 start-ups and entrepreneurs have been supported through BIG programmes. It is available to scientist entrepreneurs from research institutes, academia and start ups. The scheme is designed **to stimulate commercialisation of research discoveries by providing very early stage grants** to help bridge the gap between discovery and invention. **So, pair 1 is correctly matched.**
    - **"AmritGrand Challenge Program"** titled "जनCARE" was launched to identify **75 Start-Up Innovations in Telemedicine, Digital Health, mHealth with BIG Data, AI, BlockChain and other technologies.** It has supported 89 Digital Healthtech Innovations in Telemedicine, Digital Health, mHealth with Big Data, Artificial Intelligence (AI) Machine Learning (ML), blockchain and other technologies from Start-ups/Individuals/Companies for strengthening the Healthcare ecosystem in India. **So, pair 2 is not correctly matched.**
    - **Biotechnology Research Innovation and Entrepreneurship Development (Bio-RIDE) Scheme** with a funding support of INR 1500 Cr was approved in September 2024. It is designed **to foster innovation, promote bio-entrepreneurship, and strengthen India's position as a global leader in biomanufacturing and biotechnology**. It aims to accelerate research, enhance product development, and bridge the gap between academic research and industrial applications. The scheme would also **facilitate setting up of Biofoundry and Biomanufacturing hubs** and support start-ups. Initiatives like Biotech fund of funds-AcE under Bio-RIDE scheme would promote private investments. **So, pair 3 is not correctly matched.**

Therefore, option (a) is the correct answer.
Relevance: As per the India BioEconomy Report (IBER) 2024, the number of Biotech Start-ups has grown from 50 Start-ups in 2014 to 8,531 Start-ups in 2023.

Q3.
Answer: d
Explanation:

- **The Annual Groundwater Quality Report 2024** has been prepared by the Central Ground Water Board (CGWB). The report is based on the ground water sampling and analysis from 15,259 monitoring locations spread across the country. The major objective of the report is to study various water quality parameters like Electrical Conductivity (EC), Fluoride, Arsenic, heavy metals, Nitrate etc. in groundwater used for drinking and agriculture purposes.
- The Report finds that in certain States such as **Arunachal Pradesh, Mizoram, Meghalaya and Jammu and Kashmir, 100% of the water samples met the BIS standards. So, statement 1 is not correct.**

- The suitability of groundwater for irrigation in India is generally favorable, with the majority of samples exhibiting safe levels of sodium and alkalinity. **Over 81% of samples in the country met safe thresholds suitable for irrigation purposes.** The north-east region's groundwater quality falls in the excellent category for irrigation. **So, statement 2 is not correct.**
- The report says the **calcium-bicarbonate type dominates India's groundwater, followed by sodium and potassium ions**. Thus, bicarbonate and sodium dominate India's groundwater chemistry. **So, statement 3 is not correct.**

So, none of the above statements are correct.
Therefore, option (d) is the correct answer.

Knowledge Box
More on the Report: • Out of the samples tested, 19.8% have shown the presence of **Nitrate beyond permissible limit**, whereas Arsenic and Fluoride were high in 3.1% and 9.04% of the samples respectively. • The report points out the positive role of India's monsoon in recharging the groundwater and improving its quality. The seasonal trend of monsoon recharge effects was noticeable through parameters like Electrical Conductivity (EC) and fluoride, which enhance water quality. Percolation of pure rainwater reduces saltiness in groundwater and dilutes fluoride content. • The 2024 report flags **fluoride as a major groundwater contaminant,** with 9.04 percent of samples exceeding the BIS limit of 1.5 mg/L. **Rajasthan is worst affected,** with over 25 percent of wells in Barmer, Nagaur and Jaipur impacted by geological factors and over-extraction. • **Nitrate contamination affects 20 percent of groundwater samples,** exceeding the BIS limit of 45 mg/L. **Rajasthan is worst-hit. So, statement 2 is correct.** • **Chloride contamination in groundwater** varies regionally, with 3.07 percent of 15,259 samples exceeding the 1000 mg/L BIS limit during pre-monsoon 2023. **Rajasthan, Gujarat, and Haryana are most affected**, while coastal states like Tamil Nadu and Andhra Pradesh face seawater intrusion.

Relevance: The Annual Groundwater Quality Report 2024 released in December 2024.

Q4.
Answer: a
Explanation:

- Born to Bengali parents in Hyderabad in 1879, **Sarojini Naidu** was a woman of many talents. She was a prominent woman leader in India's freedom struggle and a devoted follower of Mahatma Gandhi, with whom she shared a warm camaraderie.
- Naidu held the distinction of being the **first Indian woman elected as the president of the Indian National Congress at its Kanpur session in 1925**. After independence, she also became the first governor of Uttar Pradesh.
- Above all, she is a pioneering figure in Indian English poetry. Naidu's poetry, often infused with rich imagery, emotive language, and musicality, earned her the title "Nightingale of India", bestowed upon her by Mahatma Gandhi. 'Damayante to Nala in the Hour of Exile' , **'In the Night' and 'In Gokhale's Garden**' are some of her poems.

Therefore, option (a) is the correct answer.

Q5.
Answer: d
Explanation:

- **Gully erosion** occurs when concentrated surface or subsurface runoff, sometimes combined with shallow mass movements, cuts deeply into the soil, forming erosional channels.
- Gully erosion stands out from other types of soil erosion in terms of depth and intensity of erosion and has a negative impact on soil quality. In India, gully erosion encompasses three distinct geomorphic features — gully systems, badlands and denuded hill slopes. Prolonged gully erosion produces deeply dissected landscapes known as 'badlands'.
- Contrary to the popular belief that Indian badlands represent the worst-case scenario of gully erosion, a recent report has found that **gully erosion in eastern India poses a more serious impediment to land degradation neutrality in India than the badlands of central and western India. So, statement 1 is not correct.**
    - A recent report says that 77 Indian districts are experiencing 'gully erosion'. 70 percent of which is in eastern and southern India.
    - The **maximum gully erosion occurs in the states of Jharkhand and Chhattisgarh, followed by Madhya Pradesh and Rajasthan.**
- Chambal river has caused a high level of gully erosion in Western India. Badlands along the lower Chambal valley has been recognised as one of India's worst forms of soil erosion. **So, statement 2 is correct.**

So, Statement–I is incorrect, but Statement–II is correct.
Therefore, option (d) is the correct answer.
Relevance: A recent article says that 77 Indian districts are experiencing 'gully erosion'.

Q6.
Answer: d
Explanation:

- A Generative Adversarial Network (GAN) is a deep learning architecture. It trains two neural networks to compete against each other to generate more authentic new data from a given training dataset.
- A GAN is called adversarial because it trains two different networks and pits them against each other.
- One network generates new data by taking an input data sample and modifying it as much as possible. The other network tries to predict whether the generated data output belongs in the original dataset.
- The GAN architecture has several applications across different industries:
    - Generative adversarial networks **create realistic images through text-based prompts** or by modifying existing images. They can help create realistic and immersive visual experiences in video games and digital entertainment. **So, point 1 is correct.**
    - Generative adversarial networks can **accurately guess and complete some missing information in a dataset.** For instance, you can train GAN to generate images of the surface below ground (sub-surface) by understanding the correlation between surface data and underground structures. **So, point 2 is correct.**

- **GAN can generate 3D models from 2D photos or scanned images.** For instance, in healthcare, GAN combines X-rays and other body scans to create realistic images of organs for surgical planning and simulation. **So, point 3 is correct.**
- Using GAN, you can generate new images from an existing image database or **original music from a database of songs. So, point 4 is correct.**

Therefore, option (d) is the correct answer.

Q7.
Answer: a
Explanation:

- The **MARCOs, or Marine Commando Force of the Indian Navy**, comprise the toughest soldiers who undergo rigorous training to conduct swift and stealthy responses. **MARCOs are modeled on the US Navy SEALS. 'The few, the fearless' is the MARCOs' motto.** The force has earned recognition for its remarkable performance in Cactus, Leech, Pawan, and Cyclone operations. Notably, their valor and courage came to prominence during their involvement in the 26/11 Mumbai attacks. **So, statement 1 is correct.**
- The force has evolved from focusing on **amphibious warfare and counterterrorism to tackling a wide range of threats, including piracy, insurgency, and grey-zone operations, that increasingly define the Indo-Pacific. So, statement 2 is not correct.**
- MARCOS can operate across diverse environments, encompassing sea, air, and land. They also engage in collaborative efforts with the Indian Army to conduct operations in areas such as Kashmir.

Therefore, option (a) is the correct answer.
Relevance: A joint desert warfare exercise of the elite special forces of the three-armed forces — Para (Special Forces) of the Army, Garud commandos of the Indian Air Force, and Marcos of the Indian Navy — was conducted in the desert areas of Rajasthan.

Q8.
Answer: b
Explanation:

- **Ferrihydrite** is the precursor for **most iron oxides in soil**, and it subsequently transforms to more stable phases such as goethite.
- The persistence of ferrihydrite in Martian dust suggests it formed during **a cold, wet period on early Mars,** followed by a transition to the **current hyper-arid environment**. **So, point 1 is correct.**
- Previous studies attributed Mars' red color to **anhydrous hematite** formed through recent weathering. Here, the recent report has shown that poorly crystalline **ferrihydrite ($Fe_5O_8H \cdot nH_2O$)** is the dominant iron oxide-bearing phase in Martian dust, based on combined analyses of orbital, in-situ, and laboratory visible near-infrared spectra.
- A widely used mineralogical model proposed that ferrihydrite in Martian dust formed by continuous oxidation and weathering under **water-poor surface conditions during the Amazonian period**, which spans from approximately 3 billion years ago.

Therefore, option (b) is the correct answer.
Relevance: Results from a new study published in Nature Communications and led by researchers from Brown University and the University of Bern suggest that the water-rich iron mineral ferrihydrite may be the main culprit behind Mars' reddish dust.

Q9.

Answer: d

Explanation:

- **Gum Arabic (GA)** or acacia gum is the exudate from the *Acacia senegal* and *Acacia seyal* trees, belonging to the Leguminosae family. It is a complex, branched heteropolysaccharide, either neutral or slightly acidic and composed of 1, 3-linked β-d-galactopyranosyl units. Most of the Gum Arabic is harvested from the arid lands of Sudan, Chad, Nigeria, Senegal, and Ethiopia. Sudan is the largest exporter, accounting for up to 80% of the trade, followed by Nigeria.
- The usage of GA dates back 5000 years. **GA has been regarded as a therapeutic for chronic kidney diseases in Middle Eastern countries. So, point 1 is correct.**
- It is renowned for its high content of non-viscous soluble fiber, making it a valuable asset in various industries, including pharmaceuticals and food. **With its emulsifying and stabilizing properties,** GA is often used as a suspending agent for insoluble drugs, further solidifying its significance in these sectors. **So, point 2 is correct.**
- The film-forming properties make it ideal for **confectionery coatings and glazes. So, point 3 is correct.**
- Within the cosmetics sector,GA serves as a stabilizer in lotions and protective creams, enhancing viscosity, facilitating spreading, and delivering a protective coating and a smooth texture**. It finds application as an adhesive agent in blusher and as a foam stabilizer in liquid soaps. So, point 4 is correct.**

Therefore, option (d) is the correct answer.

Relevance: The RSF militia, which gained control over the Gum Arabic production regions during the Sudanese civil war, is in Southern Sudan.

Q10.

Answer: a

Explanation:

- **The cat qubit technology** seeks to address one of the biggest stumbling blocks to the development of **quantum computers - making them error-free**.
- **Cat qubits are superconducting qubits** that offer an unequal trade-off in bit flip and phase flip error rates.
- While superconducting qubits commonly experience both bit flip and phase flip errors at high rates, cat qubits trade off an exponential decrease in bit flip error rates and a linear increase in phase flip error rates. The significant benefit to bit flip error rates is perceived as justifying the modest penalty in phase flip error rates.

Therefore, option (a) is the correct answer.

Relevance: Amazon has unveiled Ocelot, a prototype chip built on "cat qubit" technology - an approach derived from the famous "Schrödinger's cat" thought experiment.

Q11.
Answer: a
Explanation:

- **The TROPEX exercise** is conducted biennially, with participation by **all operational Indian Naval units along with the substantial involvement from the Indian Army, Indian Air Force, and Coast Guard assets**. TROPEX 25 is aimed at validating the Indian Navy's core warfighting skills and ensuring a synchronised, integrated response to preserve and protect national maritime security interests in a contested maritime environment against conventional, asymmetric, and hybrid threats. **So, pair 1 is correctly matched.**
- The exercise **Desert Hunt involved elite Para (Special Forces) from the Indian Army, the Marine Commandos from the Indian Navy, along with the Garud (Special Forces) from the Indian Air Force,** participating together in a simulated combat environment. This high-intensity drill enhanced interoperability, coordination, and synergy among the three Special Forces units, ensuring a swift and effective response to emerging security challenges. **So, pair 2 is correctly matched.**
- **Ekuverin is a bilateral military annual exercise conducted alternatively in India and the Maldives**. It aims to enhance interoperability in counterinsurgency and counterterrorism operations and carry out joint humanitarian assistance and disaster relief operations. **So, pair 3 is not correctly matched.**
- **Varuna is a bilateral Naval exercise between India and France, first initiated in 1993.** It has become a key pillar of Indo-French strategic cooperation**. So, pair 4 is not correctly matched.**

Therefore, option (a) is not correctly matched.

Relevance: India has launched its biggest-ever joint naval exercise with African nations, part of a push to boost its influence on the continent and presence in the Indian Ocean, where China is exercising growing commercial and military clout.

Q12.
Answer: c
Explanation:

- Recently, the Government of India launched the New Credit Assessment Model based on the scoring of digital footprints of MSMEs. It was announced in the Union Budget 2024-25 that Public sector banks (PSBs) will build their in-house capability to assess MSMEs for credit, instead of relying on **external assessment**. PSBs will develop a new credit assessment model, based on the scoring of digital footprints of MSMEs in the economy. **So, point 4 is not correct.**
- The **new credit assessment model** will leverage the digitally fetched and verifiable data available in the ecosystem and devise automated journeys for MSME Loan appraisal using objective decision-making for all loan applications and model-based limit assessment for both Existing to Bank (ETB) as well as New to Bank (NTB) MSME borrowers.
- The digital footprints used by the model may include Name and **Pan authentication using NSDL,** Mobile and email verification using OTP, **API fetch of GST data through service providers, Bank Statement Analysis using account aggregator**, ITR upload and verification, API enabled commercial and consumer bureau fetch and due diligence using CICs, fraud checks, Hunter checks through APIs, among others. **So, points 1, 2 and 3 are correct.**

- The credit assessment model for MSMEs based on digital footprints is expected to be a significant improvement over the **traditional assessment of credit eligibility based only on asset or turnover criteria.**

So, only three of the above would be considered for determining the New Credit Assessment Model for MSMEs.

Therefore, option (c) is the correct answer.

Relevance: Union Finance and Corporate Affairs Minister launched the New Credit Assessment Model based on the scoring of digital footprints of MSMEs.

Q13.

Answer: b

Explanation:

- The world's first government-funded multimodal Large Language Model (LLM) initiative, **BharatGen was launched in 2024**. It aims to enhance public service delivery and citizen engagement through foundational models in language, speech, and computer vision. BharatGen involves a consortium of AI researchers from premier academic institutions in India. **So, pair 1 is correctly matched.**
- A large language model optimised for Indian languages, **Sarvam-1 has 2 billion parameters and supports ten major Indian languages**. It is designed for applications such as language translation, text summarisation, and content generation. **So, pair 2 is not correctly matched.**
- **An open-source video transcreation platform** developed by AI4Bhārat**, Chitralekha** enables users to generate and edit audio transcripts in various Indic languages. **So, pair 3 is correctly matched.**
- A multilingual AI system, **Hanooman Everest 1.0, supports 35 Indian languages, with plans to expand to 90. So, pair 4 is not correctly matched.**

So, only two of the pairs given above are correctly matched.

Therefore, option (b) is the correct answer.

Relevance: With the approval of the IndiaAI Mission in 2024, the government allocated ₹10,300 crore over five years to strengthen AI capabilities.

Q14.

Answer: d

Explanation:

- India's goal of achieving **500 GW of renewable energy capacity by 2030** demands an unprecedented scale of financial mobilization—estimated at over ₹33 lakh crore. While the policy push for clean energy is strong, the challenge lies in bridging the massive financing gap.
- **Renewable Energy Financing Obligation** (REFO) seeks to direct **5–15% of the annual lending** by banks, NBFCs, and institutional investors toward strategic renewable energy sectors. This approach is seen as **essential to mobilising private capital** and unlocking dormant financial capacity. By mandating lending, REFO aims to ensure a consistent and affordable stream of capital for projects such as **solar parks, offshore wind, rooftop solar, PM-KUSUM, and green hydrogen**. This allocation could lower interest rates from the current 9–12% range to around **7–8%**, making clean energy **more competitive than fossil fuels**. **So, statement 1 is correct.**

- The **Renewable Purchase Obligation (RPO)** requires power distribution companies (DISCOMs), open access consumers, and captive power producers to purchase a fixed percentage of their electricity from renewable sources. In contrast, **REFO does not aim to replace RPO** but rather **complements** it by **shifting the focus from energy procurement to financing**. The two mechanisms serve different purposes and can **coexist** as part of a broader clean energy strategy. **So, statement 2 is correct.**
- India's transition to a green economy cannot rely solely on public subsidies due to limited fiscal space. REFO aims to **shift the financing burden away from the government** by mandating **private banks, NBFCs, and institutional investors** to finance renewable projects. This can unlock **trillions of rupees in private capital**, make projects more bankable, reduce reliance on public funds, and **stimulate market-led growth** in emerging sectors like **offshore wind, battery storage, and green hydrogen**. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Relevance: The Government is considering the Renewable Energy Financing Obligation (REFO), for financing renewable energy projects.

Q15.

Answer: d

Explanation:

- As part of global efforts to monitor progress toward the **Sustainable Development Goals (SDGs)**, especially **Goal 2 – Zero Hunger**, the **United Nations Statistical Commission** recently adopted a new nutrition indicator known as the **Minimum Dietary Diversity (MDD)**. The indicator would be managed jointly by the Food and Agriculture Organization of the United Nations (**UNFAO**) and the United Nations Children's Fund (**UNICEF**).
- This indicator is designed to **move beyond simple calorie counting** and emphasize the **quality and diversity of food intake**, which is critical for health, growth, development, and overall well-being.
- It is a **yes/no indicator** that determines whether **children and women of reproductive age** consumed **at least five out of ten defined food groups** in the **past 24 hours**. These food groups include grains; white roots and tubers, and plantains; pulses (beans, peas and lentils); nuts and seeds; milk and milk products; meat, poultry and fish; eggs; dark green leafy vegetables; other vitamin A-rich fruits and vegetables; other vegetables; and other fruits—chosen to reflect a well-rounded diet necessary for **adequate micronutrient intake** and overall **health and development**.
- MDD helps governments and international organizations design **evidence-based nutrition interventions**, track progress on **SDG 2**, and promote **healthy food systems**.

Therefore, option (d) is the correct answer.

Relevance: A new indicator on Minimum Dietary Diversity has been adopted by the United Nations Statistical Commission.

Q16.
Answer: a
Explanation:

- **Cassava**, also known as **'yuca' or 'manioc'**, is a tropical root crop that plays a vital role in global food security. It is particularly significant in the **tropical regions of Africa, South America, and Southeast Asia**, where it serves as a **major source of carbohydrates**. It is highly valued for its **energy-rich starchy roots**, which form a major component of daily diets, especially in **Africa, Latin America, and parts of Asia**. Due to its widespread consumption and ability to grow in marginal conditions, cassava has earned the nickname **"bread of the tropics." So, statements 1 and 2 are correct.**
- Cassava is well-known for its adaptability to a range of soil types, but it **grows particularly well in well-drained red lateritic and loamy soils**. These soils are **commonly found in tropical regions** and provide good aeration and moisture retention—conditions that support root development. **So, statement 3 is correct**.
- Cassava is **not primarily propagated through seeds**, but rather through **stem cuttings**. Farmers cut mature stems into sections, which are then planted directly into the soil. These clonal cuttings produce genetically identical plants, a method that allows quick multiplication. While cassava can reproduce sexually through seeds, this is rare in typical agricultural practice.

So, both Statement–II and Statement–III are correct and both of them explain Statement–I.
Therefore, option (a) is the correct answer.
Relevance: Thousands of years of indigenous farming myths and cultural traditions have been key to the survival of the cassava crop (the source of tapioca), a new study has found.

Q17.
Answer: b
Explanation:

- **Sal (*Shorea robusta*)** is the second most important tree after Teak in the tropical deciduous forests.
- Sal trees are venerated in Adivasi tradition. They are seen as the abode of Sarna Maa, the deity protecting the village from inclement natural forces.
- **Sarhul, literally "worship of the Sal tree",** is among the most revered Adivasi festivals. It is rooted in **nature worship** and celebrates the symbolic union of the Sun and the Earth.
- Sarhul has been celebrated by tribes such as the Oraon, Munda, Santal, Khadia, and Ho residing in the Chotanagpur plateau, with unique names for the festival and particular ways of celebrations.
- Sarhul evolved from being a hunting-centred tradition to one that revolves around agricultural processes.
- During the 19th and early 20th centuries, when tribes such as the Munda, Oraon, and Santal were sent to faraway places as indentured labour, Sarhul traveled with them. Today, the festival is celebrated in locations ranging from the tea gardens of Assam to the Andaman and Nicobar Islands, Nepal, Bangladesh, and Bhutan.

Therefore, option (b) is the correct answer.

Q18.
Answer: b
Explanation:

- **Sea grasses** are a unique and ecologically vital group of marine **angiosperms (flowering plants)** that thrive in shallow, coastal saltwater environments. Despite their grass-like appearance, they are not true grasses but **fully developed higher plants** that possess true roots, stems, and leaves.
- Sea grasses are **not algae**, but **true flowering plants (angiosperms)**. Unlike algae, which lack differentiated structures, sea grasses possess **true roots, rhizomes, and strap-like or oval leaves**. These features allow them to anchor into sandy or muddy sea floors, absorb nutrients from the substrate, and perform photosynthesis efficiently in marine environments. In contrast, **macroscopic algae** (like seaweed) do not have vascular tissue or true roots, stems, and leaves. **So, statement 1 is not correct.**
- India's **richest sea grass meadows** are concentrated along the **southeast coast of Tamil Nadu**, particularly around the **Gulf of Mannar and Palk Bay**. These areas offer ideal conditions for sea grass growth—**high salinity, clear waters, and sandy substrates**. Additionally, significant sea grass populations are also found in **lagoons of some Lakshadweep Islands** and around parts of the **Andaman and Nicobar Islands**. These ecosystems support diverse marine life and contribute to the ecological and economic health of coastal communities. **So, statement 2 is correct.**
- **Agar-agar and iodine** products are actually derived from **marine macroalgae seaweeds**. Sea grasses are **not used for commercial extraction** of such products. **So, statement 3 is not correct.**

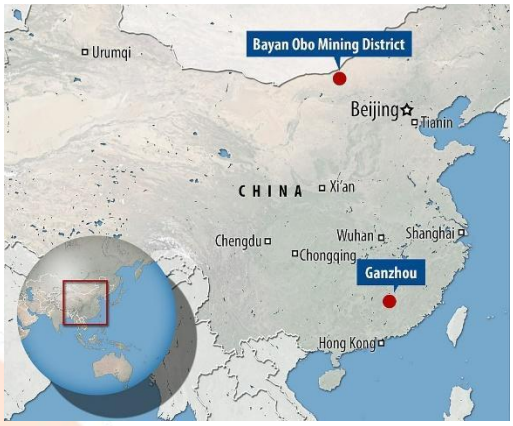
Therefore, option (b) is the correct answer.
Relevance: According to recent study, seagrass has been declining at a rate of 1-2 per cent per year for the past century and nearly 5 percent of species are now endangered.

Q19.
Answer: c
Explanation:

- **The Jaipur Declaration (2025–34) and Hanoi Declaration (2013–23) aim to promote 3R (Reduce, Reuse, Recycle) principles and facilitate the transition towards a circular economy in the Asia-Pacific region.** These declarations provide a voluntary, legally non-binding framework for countries to shift from a linear economy to a more sustainable and resource-efficient model.
- The declarations are part of the Regional 3R and Circular Economy Forum, launched by UNCRD in 2009, to offer strategic guidance to governments on integrating circularity in policy and practice. India will host the 12th Regional Forum in Jaipur (2025), where the Jaipur Declaration will be adopted.
- The **Jaipur Declaration builds upon the Hanoi Declaration**, expanding its scope to align with the **Sustainable Development Goals (SDGs)** and **carbon neutrality objectives** for the coming decade.

Therefore, option (c) is the correct answer.
Relevance: The 12th Regional 3R and Circular Economy Forum was held in Jaipur in March 2025.

Q20.
Answer: b
Explanation:



- **Bayan Obo mining complex in Inner Mongolia, China, has recently been identified as a significant source of thorium.** A declassified geological survey estimates the region could yield **over 1 million tonnes of thorium**, making it geopolitically and strategically important in the evolving nuclear energy landscape.
- **Thorium is a naturally occurring radioactive element** that is more abundant in the Earth's crust than uranium. It is mainly found in the mineral **monazite**, which also contains rare earth elements.
- **Thorium-232**, the naturally occurring isotope, is **not fissile by itself** but can be converted into **Uranium-233**, a fissile material, through irradiation. This makes thorium a potential **alternative nuclear fuel**, especially in **advanced reactors** such as **PHWRs, PWRs, and FNRs**.
- While India holds the **world's largest thorium reserves**, the new findings from Bayan Obo could alter **strategic balances in future nuclear fuel sourcing**, especially as the world looks for **low-carbon energy alternatives**.

Therefore, option (b) is the correct answer.
Relevance: A 2025 declassified geological report in China revealed that the Bayan Obo mining complex in Inner Mongolia may contain 1 million tonnes of thorium.

Q21.
Answer: c
Explanation:

- **The Convention on Cluster Munitions (CCM)** is a **multilateral treaty** that seeks to eliminate the use and impact of cluster munitions worldwide. These weapons scatter submunitions ("bomblets") over wide areas, posing long-term dangers to civilians, particularly in post-conflict zones. The Convention represents a significant milestone in humanitarian disarmament efforts.
- The CCM was adopted on 30 May 2008 in Dublin, signed on 3 December 2008 in Oslo, and entered into force on 1 August 2010. As of 2025, it has 112 States Parties and 12 signatories yet to ratify.
- **The Convention prohibits all use, production, transfer, and stockpiling of cluster munitions.** It also obliges States Parties to provide victim assistance, clear contaminated areas, and destroy existing stockpiles. **So, statement 1 is correct**.
- **The Convention does not have a legally binding enforcement mechanism** under the United Nations or any other global enforcement body. Compliance is **voluntary** and promoted through **peer pressure, civil society advocacy, and international cooperation**. **So, statement 2 is not correct**.

- **India, the United States, Russia, China, Ukraine, and Israel have not signed the Convention** due to strategic and military reasons. These non-signatories argue that cluster munitions have tactical value in warfare. **So, statement 3 is correct**.
- In **March 2025**, **Lithuania became the first country to withdraw** from the Convention, citing regional security concerns over Russia. This move was condemned by humanitarian groups and raised alarms over the weakening of global disarmament norms. **So, statement 4 is correct**.

Therefore, option (c) is the correct answer.

Relevance: Lithuania's withdrawal from the Convention on Cluster Munitions (CCM) in March 2025 marks a turning point in disarmament diplomacy.

Q22.

Answer: c

Explanation:

- **Anti-competitive pricing** refers to pricing strategies used to distort fair market competition, often by dominant players to suppress or eliminate rivals. The **Competition Commission of India (CCI)** is empowered under the **Competition Act, 2002** to regulate such practices, especially those that adversely impact consumer welfare or harm smaller competitors.
- **The Competition Act, 2002** prohibits certain anti-competitive practices including **predatory pricing** under Section 4(2)(a)(ii). Predatory pricing is defined as the practice where a **dominant firm** sells goods or services **below cost of production** to **eliminate competition**. **So, pair 1 is correctly matched.**
- **Limit Pricing** is a strategy where an incumbent firm sets **prices low enough to deter entry** of new firms into the market. It does not necessarily involve selling below cost rice, but it's still an **entry barrier tactic**, hence considered anti-competitive in certain contexts. **So, pair 2 is correctly matched.**
- **Collusive Pricing** does not mean offering the **lowest prices to consumers**, but refers to **price-fixing agreements between competitors** to reduce competition and control the market. This harms consumers and is prohibited under **Section 3** of the Competition Act. **So, pair 3 is not correctly matched.**
- **Price Skimming** is a **legitimate pricing strategy**. It involves setting **initial high prices** for a new product to recover R&D costs and target early adopters. It is not associated with harming competition. **So, pair 4 is correctly matched.**

So, only three of the pairs given above are correctly matched.

Therefore, option (c) is the correct answer.

Knowledge Box

Competition Commission of India:
• The **Competition Commission of India (CCI)** was constituted in **2003** and became fully operational in **2009** under the **Competition Act, 2002**. It functions as a **statutory body under the Ministry of Corporate Affairs**.
• The CCI's mandate includes:
  o Preventing anti-competitive agreements (Section 3)
  o Regulating abuse of dominance (Section 4)
  o Overseeing mergers and acquisitions (Sections 5 and 6)

- Conducting investigations against unfair trade practices (Section 19)
- Promoting consumer welfare and economic efficiency

- The CCI is composed of a **Chairperson and up to six members**, appointed by the central government. Members must have at least **15 years of professional experience** in law, economics, finance, or public affairs.
- CCI replaced the outdated **MRTP Act, 1969**, and follows global best practices recommended by the **Raghavan Committee**. It plays a key role in ensuring fair market conduct in India's liberalized economy.

Relevant: Recently, the Competition Commission of India (CCI) released draft regulations titled The Competition Commission of India (Determination of Cost of Production) Regulations, 2025.

Q23.
Answer: b
Explanation:

- **Perfluorosulfonic acid (PFSA) polymers**, such as **Nafion**, are widely used as the **electrolyte membrane** in **Proton Exchange Membrane Fuel Cells (PEMFCs)**.
- These polymers have **excellent proton conductivity**, allowing **hydrogen ions (protons)** to pass from the anode to the cathode, which is essential for maintaining the electrochemical reaction in the fuel cell.
- At the same time, PFSA polymers are **electrically insulating**, meaning they do **not conduct electrons**, which is critical to ensuring that electrons flow through the external circuit, generating electric current.
- The membrane also acts as a **barrier to gases**, preventing hydrogen and oxygen from directly mixing within the cell, which enhances **efficiency and safety**.
- Thus, perfluorosulfonic acid polymers serve the **dual purpose** of **proton exchange** and **electrical insulation**, making them ideal for use in PEMFCs.

Therefore, option (b) is the correct answer.
Relevance: Proton Exchange Membrane Fuel Cells are increasingly used in clean energy systems, especially in automobiles, portable power devices, and backup systems.

Q24.
Answer: d
Explanation:

- **Solar Storm** is a sudden explosion of particles, energy, magnetic fields, and material blasted into the solar system by the Sun.
- The Sun creates a **tangled mess of magnetic fields** — kind of like a disheveled head of hair after a fitful night of sleep. These magnetic fields get twisted up as the Sun rotates — with its equator rotating faster than its poles. Solar storms typically begin when these twisted magnetic fields on the Sun get contorted and stretched so much that they snap and reconnect (in a process called magnetic reconnection), releasing large amounts of energy. **So, statement 1 is correct.**
- Solar Storm can generate **Solar Flare** (bright flash of light), **Radiation Storm** (flurry of solar particles propelled into space at high speeds) and **Coronal Mass Ejection** (an enormous cloud of solar material).

- When directed toward Earth, a solar storm can create a major disturbance in Earth's magnetic field, called a **Geomagnetic Storm**, that can produce effects such as radio blackouts, power outages, and beautiful auroras. They do not cause direct harm to organisms on Earth; however, Earth's magnetic field and atmosphere protect us from the worst of these storms. **So, statement 2 is correct.**
- Aditya-L1, India's first dedicated space based solar mission, has made a ground-breaking observation from its scientific payloads- **capturing the first-ever image of a solar flare 'Kernel' in the lower solar atmosphere,** namely the photosphere and the chromosphere, in the images recorded in the Near Ultraviolet (NUV) band. **So, statement 3 is correct.**
    - **The Aditya-L1 mission** was launched on September 2, 2023 by Indian Space Research Organisation (ISRO) PSLV C-57 rocket.
- Solar flares impact Earth only when they take place on the side of the Sun that is facing Earth. Solar flares are rated into different classes based on their strength, or energy output, and the effect a flare will have on Earth depends on what class it is (B, C, M, and X classes, with X being the most intense).

Therefore, option (d) is the correct answer.

Relevance: The Sun recently unleashed a powerful X1.2-class solar flare, which is one of the strongest types of solar flares.

Q25.

Answer: b

Explanation:

- In the **World Development Indicators** (the primary World Bank collection of development indicators) all 189 World Bank member countries, along with 28 others with populations of more than 30,000 are **classified by income level** and geographic region for the presentation of key statistics allowing users to aggregate, group, and compare statistical data as needed.
- The world's Middle Income Countries (MICs) are a diverse group by size, population, and income level. On the basis of income levels, these economies are currently divided into four income groupings: Low-income economies, Lower middle-income economies, Upper middle-income economies, and High-income economies. **Income is measured using Gross National Income (GNI) per capita, in U.S. dollars,** converted from local currency using the **World Bank Atlas method.**
- For cross-national comparisons, **estimates are converted from local currency units (LCU) to current U.S. dollars using the Atlas method,** referring to a former World Bank publication called the Atlas of Global Development. The Atlas method smooths exchange rate fluctuations using a three-year moving average, price-adjusted conversion factor. The USD estimate of GNI per capita is derived by applying the Atlas conversion factor to estimates measured in LCU.
- For the 2024 fiscal year, **Low-income economies** are defined as those with a GNI per capita, calculated using the **World Bank Atlas method**, of **$1,135 or less in 2022. Lower middle-income economies** are those with a GNI per capita **between $1,136 and $4,465; Upper middle-income economies** are those with a GNI per capita **between $4,466 and $13,845**; **High-income economies** are those with a GNI per capita of **$13,846 or more,** according to the World Bank Country and Lending Groups.

- MICs are home to **75% of the world's population and 62% of the world's poor.** At the same time, MICs represent about **one third of global GDP and are major engines of global growth.** The World Bank Group continues to evolve its partnership with MICs, working with them simultaneously as clients, shareholders, and global actors.

Therefore, option (b) is the answer.

Relevance: Recently published IMF-World Bank report states that current laws allow the government to control senior management and boards of regulators.

Q26.

Answer: c

Explanation:

- **India-EU Trade and Technology Council (TTC)** was established in February 2023 by India and the European Union as a key bilateral platform to address challenges at the confluence of trade, trusted technology and security.
- **This TTC is the first of its kind for India** and the second for the European Union (EU), as the EU's first was launched with the United States (US) in 2021. **So, statement 1 is correct.**
- It is a key forum to deepen the strategic partnership on trade and technology between the two partners. Geostrategic challenges have reinforced the EU and India's common interest in ensuring security, prosperity and sustainable development based on shared values. The TTC will help increase EU-India bilateral trade, which is at historical highs
- It has a Working Group on **Strategic Technologies, Digital Governance, and Digital Connectivity** which functions on cooperation on data governance and human-centric digitalization. This covers research, rules and regulation, and joint scientific and technical capacity building in the following areas of mutual interest: 5G/6G, AI, semiconductors, etc. This serves as an overarching platform to exchange views on the strategic, political and security implications of **new and emerging technologies, and discuss critical and emerging technology standards and certification. So, statement 3 is correct.**
- The TTC has **not replaced the India-EU Free Trade Agreement**. Also, the work under Working Group 3 complements the ongoing negotiations on a Free Trade Agreement (FTA), an Investment Protection Agreement (IPA) and a Geographical Indications Agreement which are proceeding on **separate tracks. So, statement 2 is not correct.**

Therefore, option (c) is the correct answer.

Relevance: The second meeting of the India-EU Trade and Technology Council (TTC) was recently taking place in New Delhi.

Q27.

Answer: b

Explanation:

- The **Asiatic Lion**, endemic to Gir landscape of Gujarat, is listed in Schedule-I of Wildlife (Protection) Act, 1972 and has accorded the highest degree of protection. Conservation efforts in this region have seen the lion population grow from around 523 in 2015 to approximately 674 in 2020, a testament to the success of dedicated protection measures.

- **Greater Gir Concept (GGC)** involves developing additional suitable habitats for lions beyond the traditional Gir National Park & Sanctuary. Sanctuaries like **Girnar, Pania, and Mitiyala** have been notified to expand the protected area network. **So, points 2, 3 and 5 are correct.**
- **Ratanmahal Sloth Bear Sanctuary and Nal Sarovar Bird Sanctuary** have **not** been notified as parts of the GGC. **So, points 1 and 4 are not correct.**

Therefore, option (b) is the correct answer.

Relevance: The Forest Minister recently informed the Gujarat Legislative Assembly about the deaths of the lions and leopards reported over the past two years.

Q28.

Answer: a

Explanation:

- According to the **India Philanthropy Report 2025** by Bain & Company and Dasra, India's overall social sector funding grew at approximately 13% over the past five years, **dominated by the public sector.** By the end of FY 2024, it is estimated to have reached ~INR 25 lakh crore ($300 billion; 8.3% of GDP), with public spending accounting for 95% of total public expenditure. **So, statement 1 is correct.**
- **Family-owned/run businesses contribute 65%–70% of private-sector Corporate Social Responsibility (CSR) spending annually**, totaling approximately INR 18,000 crore ($2.2 billion), with the top 2% of family-owned/run firms contributing 50%–55% of the total family-owned/run businesses' CSR contribution, highlighting the outsized role of a few key players. **So, statement 2 is correct.**
- Private sector funding has been **growing at 9% annually in the last five years,** driven by domestic giving across both family philanthropy and CSR spending. Public sector spending has **been growing at 13% annually in the last five years** driven by spending toward achieving Sustainable Development Goals (SDGs) 2030 across traditional (healthcare, education) and emerging causes (climate action, gender equality). **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.

Relevance: Recently, the India Philanthropy Report 2025 was published by Bain & Company and Dasra.

Q29.

Answer: c

Explanation:

- **Antarctic Circumpolar Current (ACC)** is the strongest ocean current on the planet. It forms part of the global ocean "**conveyor belt**" connecting the Pacific, Atlantic, and Indian oceans. The system regulates Earth's climate and pumps water, heat and nutrients around the globe.
- **Weakening of the ACC could aid the entry of invasive species** such as southern bull kelp to Antarctica, disrupting local ecosystems and food webs. **So, point 1 is correct.**
- A weaker ACC may also allow more warm water to penetrate southwards, exacerbating the **melting of Antarctic ice shelves and contributing to global sea-level rise**. Faster ice melting could lead to further weakening of the current, commencing a vicious spiral of the current slowdown. **So, point 2 is correct.**

- Weakening of the ACC could **reduce biodiversity and decrease the productivity of fisheries** that many coastal communities rely on. **So, point 3 is not correct and point 4 is correct.**

Therefore, option (c) is the correct answer.

Relevance: New research using a global ocean simulation model shows that by 2050, the strength of the Antarctic Circumpolar Current will decline by ~20% for a high-emissions scenario.

Q30.

Answer: b

Explanation:

- **Assam Rifles** is one of the six Central Armed Police Forces (CAPFs) that come under the MHA's administrative control.
- **Assam Rifles** has its origins in 1835 as a militia called the **'Cachar Levy**' to protect British tea estates and their settlements against tribal raids.
    - It later became the 'Frontier Force' conducting punitive expeditions across the borders of Assam.
    - In 1870, the existing elements were merged into three Assam Military Police Battalions, sending over 3,000 men as part of the British Army to Europe and West Asia. In 1917, the name was changed to Assam Rifles.
    - The force has evolved post-Independence, from combat roles during the Sino-India War 1962 to being a part of the Indian Peace Keeping Force (IPKF) in Sri Lanka in 1987, besides performing peacekeeping roles in north-eastern India.
- It conducts **counterinsurgency operations** in north-eastern India and other areas where deemed necessary.
- Its operational control is with the Indian Army, with an officer of the rank of **Lieutenant General** commanding the force.
- The headquarters of the Director-General of Assam Rifles (DGAR) is in **Shillong** unlike other Central paramilitary forces, all of which have their headquarters in New Delhi.

Therefore, option (b) is the correct answer.

Relevance: Union Home Minister Amit Shah recently announced the relocation of Assam Rifles battalions to Mizoram.

Q31.

Answer: a

Explanation:

- **Space anaemia** is a condition observed in astronauts during and after extended space missions. Unlike anaemia on Earth, which is often caused by nutritional deficiencies or chronic diseases, space anaemia is primarily due to the body's adaptation to the unique environment of space. **So, statement 1 is correct.**
- In microgravity, the body experiences significant fluid shifts. On Earth, gravity helps maintain the distribution of blood and fluids throughout the body. However, in space, this **gravitational pull is absent**, leading to an upward shift of bodily fluids towards the head. This shift **impacts blood volume and the production of red blood cells.** Space anaemia arises as a consequence of these fluid shifts. **So, statement 2 is correct.**

So, both Statement-I and Statement-II are correct and Statement-II explains Statement-I. Therefore, option (a) is the correct answer.

Relevance: Astronaut Sunita Williams' concerns about the health effects of space travel have gained attention recently.

Q32.

Answer: d

Explanation:

- The **Mascarenes** are an island group lying near the Tropic of Capricorn in the southwestern Indian Ocean ~700 km east of Madagascar.
- This archipelago comprises three high volcanic islands (**Réunion, Mauritius, Rodrigues**), scattered along a ~600 km west-east axis, and a group of **small coralline islands** (**Cargados** Carajos Shoals) ~400 km to the north of Mauritius, which sit upon a submarine bank of volcanic origin that extends a further 700 km or more to the northeast. **So, points 1, 3 and 4 are correct.**
- **Minicoy Island is not part of the Mascarene Islands.** It is a part of the Lakshadweep archipelago. **So, point 2 is not correct.**

Therefore, option (d) is the correct answer.

Knowledge Box
Archipelago: ● Archipelago is a group or chain of islands closely scattered in a body of water, typically the ocean, but sometimes also a lake or river. ● Most archipelagos are formed when volcanoes erupt from the ocean floor; these are called oceanic islands. ● It can also form as a result of erosion, sedimentary deposits, rising sea level, and other geographic processes.

Q33.

Answer: b

Explanation:

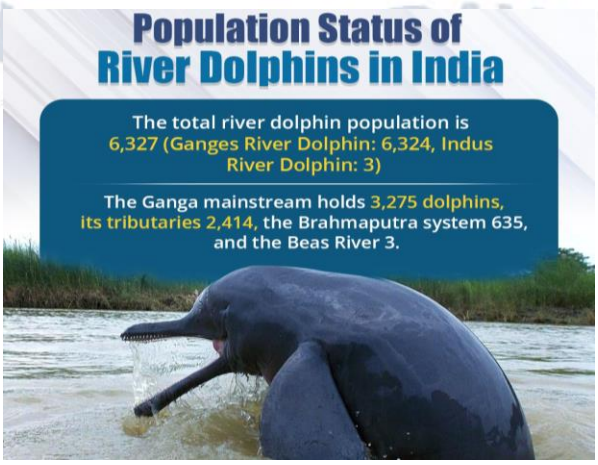
- **"Prithvi Vigyan (PRITHVI)"** is an umbrella scheme by merging the existing five schemes "Atmosphere & Climate Research-Modelling Observing Systems & Services (ACROSS)", "Ocean Services, Modelling Application, Resources and Technology (O-SMART)", "Polar Science and Cryosphere Research (PACER)", "Seismology and Geosciences (SAGE)" and "Research, Education, Training and Outreach (REACHOUT)".
- There are four sub-schemes under the PRITHVI umbrella scheme that are implemented by various institutions under the MoES. The sub-schemes along with the unit implementing them are as follows:
    - **ACROSS**, its objective to deploy and sustain the state-of-the-art observing systems to monitor weather, climate and ocean parameters. **So, pair 1 is correctly matched.**
    - **OSMART** is being implemented by National Institute of Ocean Technology (NIOT), Indian National Centre for Ocean Information and Services (INCOIS), Centre for Marine Living Resources and Ecology (CMLRE) and National Centre for Coastal Research (NCCR).

- The objective is to deploy and sustain a wide range of state-of-the-art ocean observation systems for the acquisition of real-time data from the seas around India.
- Generate and regularly update information on **marine living resources** and their relationship with the physical environment in the **Indian Exclusive Economic Zone** (EEZ). **So, pair 2 is correctly matched.**

- **PACER** is being implemented by the National Centre for Polar and Ocean Research (NCPOR).
  - The objective is to maintain **Indian Research bases at Antarctica, Arctic and Himalaya**, establishment of a state-of-the-art polar research and logistic facilities in the country. **So, pair 3 is correctly matched.**
- **SAGE** is being implemented by the National Centre for Seismology (NCS) and National Centre for Earth Sciences Studies (NCESS).
  - The objectives of SAGE are to **monitor earthquake activity** in and around the country on a 24x7 basis and preparation of earthquake hazard related products for selected cities in India. **So, pair 4 is not correctly matched.**
  - Deep Borehole Observatory for directly measuring the **in-situ physical properties of rocks, pore fluid pressure**, hydrology parameters, temperature and other parameters of an intra-plate, active fault zone in the near–field of earthquake-before, during and after their occurrence.
- **REACHOUT**, the objective to support various **R &D activities** in the thrust areas of different components of Earth System Sciences that are theme and need-based, and that would help in attaining the National goals set up for MoES. **So, pair 5 is not correctly matched.**

So, only three of the above pairs are correctly matched.
Therefore, option (b) is the correct answer.
Relevance: The Union Cabinet recently approved the overarching scheme "PRITHVI VIgyan (PRITHVI)" of the Ministry of Earth Sciences.

Q34.
Answer: b
Explanation:



- The first-ever comprehensive survey of river dolphins has revealed a population estimate of **6,327 dolphins in the Ganga and Brahmaputra basins and 3 in the Indus basin.** The survey, conducted under Project Dolphin launched in 2020.

- India is home to two species of freshwater dolphins — **Ganges (*Platanista gangetica*) and Indus (*Platanista minor*)**, which **s**erve as bio-indicators of healthy river ecosystems.
- Uttar Pradesh has the highest number of dolphins at 2,397, followed by Bihar 2,220, West Bengal 815, Assam 635, Jharkhand 162, Rajasthan and Madhya Pradesh 95, and Punjab 3, the report said.
- The Ganges river dolphin, known for its unique features, is found in the Ganga-Brahmaputra-Meghna river system and its tributaries across India, Bangladesh, Nepal, and Bhutan.
- The findings indicate that Assam remains one of the key conservation areas for the endangered River Dolphin and the Brahmaputra River holds a stable and barrier-free population of these dolphins, making it a critical habitat for their long-term survival.
    - However, the survey raised concerns about other river systems in Assam. The **Barak River, once home to these dolphins, recorded no sightings**, indicating possible local extinction. **So, point 1 is not correct.**
    - Additionally, the **Subansiri River has shown a rapid decline in dolphin numbers**, flagging it as an area of concern for conservationists. **So, point 4 is correct.**
- The report pointed out that the Indus River Dolphin in India is 3 and is currently observed only in the **Beas** river. **So, point 3 is correct and 2 is not correct.**
- Gangetic dolphins and Indus dolphins are classified as **endangered and listed as Schedule I** species under the provisions of the Wild Life (Protection) Act, 1972.

Therefore, option (b) is the correct answer.

Relevance: The first-ever comprehensive survey of river dolphins has recently been conducted under Project Dolphin.

Q35.

Answer: d

Explanation:

- The **Foreigners Act, 1946** (Act No. 31 of 1946), which came into effect on **23rd November 1946**, empowers the **Central Government** to regulate the entry, presence, and departure of foreigners in India.
- The act defines 'foreigner' as a person who is not a citizen of India and it empowers the Government to make provisions for prohibiting, regulating or restricting the entry of foreigners in the country. **So, statement 2 is not correct.**
- It is to be noted here that under this act, the 'burden of proof' for proving one's nationality lies with the person in question and not with the authority challenging it.
- While the Central Government is vested with the powers to detain, deport a foreign national staying in India under Section 3(2)(e) & 3(2)(c) of the Foreigner's act, 1946 which is a pre-constitution act.
- These powers were later extended to the State Governments and Union Territory Administration as well under Articles 258(1) & 239(1) of the Constitution respectively. Therefore, while the Central Government has a **statutory mandate** to detain and deport a foreigner, the State Government has a constitutional mandate for the same. **So, statement 1 is not correct.**

Therefore, option (d) is the correct answer.

Relevance: The Central Government has proposed the Immigrants and Foreigners Bill, 2025, in Parliament. This bill (when passed) will repeal the Foreigners Act, 1946, along with three other laws.

Q36.
Answer: c
Explanation:

- The **Unlawful Activities (Prevention) Act (UAPA)** was enacted in 1967 to address anti-national activities and secessionist movements threatening India's sovereignty and territorial integrity.
- Under the Act, the Central Government is **empowered to designate any organization or individual as terrorist organization or terrorist respectively**, the latter power was added only via the latest amendment to the act in 2019. **So, statement 1 is correct**.
- The Act **doesn't define the words 'terror' or 'terrorist'** but **defines the term 'terrorist acts**', at the same time it also explicitly defines '**unlawful activity'** to include 'acts or words that support cession, secession, or disrupt India's sovereignty and integrity. **So, statement 2 is correct**.
- If any organization is declared unlawful under the Act, the decision automatically gets referred to a tribunal which conducts a judicial enquiry giving the now banned organization a chance to present its case. **So, statement 3 is correct**.

So, all three of the above statements are correct.
Therefore, option (c) is the correct answer.
Relevance: The Central Government has declared two Hurriyat groups in J&K unlawful under UAPA.

Q37.
Answer: b
Explanation:

- **Mauritius** was a former colony of France and Britain, gaining independence in 1968. During **French rule**, the **Indians** were brought here in **1729** from today's **Puducherry region** to work as **artisans and masons**. Later, during the British rule over the island nation nearly **500,000 Indian indentured labourers** were brought to Mauritius to work on **sugarcane plantations**. The **first batch of 36 workers** arrived on **November 2, 1834**, aboard the ship **'Atlas'**. This date is commemorated as **'Aapravasi Diwas'** in Mauritius.
- **Mahatma Gandhi visited Mauritius** from **October 29 to November 15, 1901**, on his return journey from South Africa. During his visit, Gandhi delivered three key messages to the Indian diaspora: the importance of education, political participation, and maintaining cultural ties with India.
- In his honour, **Mauritius celebrates its National Day on March 12**, which marks the date of Gandhi's **Dandi Salt March (1930)** in India.
- Gandhiji also revered the Mauritian lands as can be deduced from his reference to Mauritius in speeches as Congress President (1924) and again during the Partition (1947), using it as an example of communal harmony and unity among Hindus and Muslims.

Therefore, option (b) is the correct answer.
Relevance: The Indian Prime Minister was the chief guest for the national day of Mauritius in 2025.

Q38.
Answer: c
Explanation:

- **Sadeli Craft** is a **traditional wood inlay craft** (marquetry) known for its **intricate geometric and floral patterns**. It was introduced to **Surat region in Gujarat**, by the **Parsi community** from Iran. **So, statement 1 is correct.**
- It is known for its geometric patterns, resembling Persian Khatam work, and involves meticulous inlay of wood, metal, or ivory. The **Petigara families** of Surat are traditionally associated with the preservation and continuation of the craft.
- The State of **Gujarat** was recently awarded the Geographical Indication (GI) tag for the craft in 2024. **So, statement 2 is not correct.**
- The intricate inlaying and detailing process used for making these craftworks require a wood that is durable enough to endure the intense carving. One of the major reasons for the craft finding its home in India was the abundance of Teakwood which was optimum for the carving purposes. **So, statement 3 is correct**.

Therefore, option (c) is the correct answer.
Relevance: Prime Minister in his recent visit to Mauritius, presented its first lady with a Banarasi saree packed in a Sadeli box.

Q39.
Answer: d
Explanation:

- **Ethylene** is a **gaseous plant hormone** that plays a crucial role in **fruit ripening**, **flower wilting**, and **leaf fall**. In fruits like bananas, ethylene production continues even after harvest, accelerating the **ripening and browning process**. It activates several genes, notably the one responsible for the enzyme **polyphenol oxidase (PPO)**, which breaks down pigments and causes browning when exposed to oxygen.
- Physiological processes regulated by Ethylene:
  - **Fruit ripening**: Ethylene promotes the breakdown of cell walls, softening of fruit, and development of flavor and aroma.
  - **Flower senescence**: Ethylene regulates the aging process in flowers.
  - **Leaf abscission**: Ethylene helps in the shedding of leaves.
  - Ethylene is often referred to as the "**ripening hormone**" because of its role in promoting fruit ripening. **So, statement 3 is correct.**
- Polyethylene bags are known to release this gas; this is the reason that fruits kept in them ripens fast.

Therefore, option (d) is the correct answer.
Relevance: Recently, Tropic, a UK-based biotech company has developed gene edited bananas with reduced expression of PPO producing genes.

Q40.
Answer: c
Explanation:

- **Domicile** is a legal concept for the purpose of **determining which legal system or which set of laws are applicable to an individual**, even if the individual has no permanent home he/she is invested with a domicile by law.
- The Constitution of India in **Article 5 refers and mentions the 'domicile in the territory of India'** but the term **domicile is not defined** anywhere in the Constitution. **So, statement 1 is correct.**
- In India, the respective **State governments issue a domicile certificate** to the residents of the respective State, irrespective of caste, creed, language or religion. The certificate indicates that the individual is a permanent resident of that State. Women marrying permanent residents of another State can apply for a domicile certificate of that State. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.
Relevance: In a recent judgment, the Supreme Court of India declared that domicile-based reservations for postgraduate (PG) medical admissions within the State quota are unconstitutional.

Q41.
Answer: c
Explanation:

- **Kanyasulkam** (1892), is considered one of the greatest plays in the Telugu language. The play exposed the evils of child marriage and the dowry system, bringing critical social issues into the public discourse.
- The play was written by the famous poet **Gurjada Apparao**, he was a renowned Telugu playwright, poet, and social reformer. Born on September 21, 1862, in Rayavaram, Andhra Pradesh, he played a pioneering role in introducing colloquial Telugu literature.
- He famously gave the treatise ***Desamante matti kaadoyi, Desamante manushuloyi"*,** which translates to "A country is not just its soil, a country is its people".

Therefore, option (c) is the correct answer.
Relevance: The poet was recently quoted by the Union Finance Minister in her budget speech.

Q42.
Answer: c
Explanation:

- It is **Article 15** in Part III of the Constitution of India that explicitly provides that 'No citizen shall, on grounds only of religion, race, caste, sex, place of birth or any of them, be subject to any disability, liability, restriction or condition with regard to
    - **access to shops, public restaurants, hotels and places of public entertainment. So, point 2 is correct.**
    - **the use of wells, tanks, bathing ghats**, roads and places of public resort maintained wholly or partly out of State funds or dedicated to the use of the general public. **So, point 1 is correct.**
- Further, though denying **any person entry to a place of worship is the traditional form of untouchability**, it is not explicitly written in the Constitution. **So, point 3 is correct.**

So, all three of the above activities are prohibited under Article 15 of the Constitution of India.

Therefore, option (c) is the correct answer.

Relevance: Recently in March 2024, Dalits were allowed to enter the historic Gidheswara temple in West Bengal for the first time.

Q43.

Answer: d

Explanation:

- **SPHEREx** is a **NASA space telescope** that was launched in February 2025. SPHEREx stands for **Spectrophotometer for the History of Universe, Epoch of Reionization and Ice Explorer**.
- This telescope will map the universe through detection of **Infrared as well as optical wavelength radiations** – Optical light is as visible to the human eye, and was also employed on Hubble telescope, while Infrared is invisible and is more akin to heat signature. Detection of both these wavelengths empowers the telescope to create an **all-sky spectral map** with **color resolution** to help identify structures invisible in other wavelengths. **So, statement 1 is correct**.
- Its mission is to create the first-ever **full-sky spectral survey** in **infrared wavelengths**, aimed at studying the **early universe**, **galactic evolution**, and astrobiological molecules. Unlike the James Webb Space Telescope (JWST), which focuses on deep, localized observations, SPHEREx offers panoramic, **color-rich mapping of the universe**. **So, statement 2 is correct**.
- The telescope is launched into a **sun-synchronous polar orbit**, and is kept **consistently oriented away from the sun** and Earthshine to maintain the low temperatures required for infrared detection. **So, statement 3 is correct**.
    - **ADITYA L1** of ISRO is facing towards the sun, while the Parker **probe** of NASA is facing away from it.

Therefore, option (d) is the correct answer.

Q44.

Answer: c

Explanation:

- **E.V. Ramaswamy Naicker**, popularly known as Periyar, was a pioneering rationalist, social reformer, and the founder of the Self-Respect Movement in Tamil Nadu.
- Naicker was a **proponent of 'Dravidian identity'** which he adopted in antagonism to Brahmanical and Sanskrit dominance in the Indian society at large. As a part of this identity politics, **he encouraged the use of Tamil in public discourse**. He envisaged the same as a tool for dignity, equality and transformation, but was highly opposed to the notion of linguistic chauvinism. **So, statement 1 is correct.**
- He encouraged the **revival of Tamil** as a **modern and functional language**, especially for governance and public life and was the one who popularized the use of the word *Vannakam* for salutation instead of the Sanskrit word *namaskaram* in the Dravid region.
- As a rationalist and social reformer, Periyar **rejected linguistic nationalism**, caste, and religion. He was **critical of** Bhakti literature and **linguistic purism**, which he viewed as reinforcing caste and superstition**.** In fact, he **opposed the Tanittamil (Pure Tamil) movement** as overly conservative and restrictive. He believed linguistic pride alone would not uplift the oppressed and that Tamil must evolve to meet modern needs, including borrowing from other languages if necessary. **So, statement 2 is not correct.**

So, Statement-I is correct, but Statement-II is incorrect

Therefore, option (c) is the correct answer.
Relevance: Union Finance Minister recently cited Periyar's criticism of Tamil as "a barbaric language" during a debate.

Q45.
Answer: b
Explanation:

- **Houthis:** This group is officially known as *Ansar Allah* (Partisans of God), is a Shiite Muslim political and military movement based in **Yemen**. Originating in the 1990s under the leadership of Hussein al-Houthi, the group represents the interests of the Zaidi sect, a minority branch of Shiite Islam distinct from dominant Shiite groups in Iran and Iraq. **So, pair 1 is correctly matched.**
- **M23 Rebels**: It is an armed rebel group operating in the eastern **Democratic Republic of Congo (DRC)**. The group derives its name from a March 23, 2009 agreement between the Congolese government and the Tutsi-led National Congress for the Defence of the People (CNDP), which promised the integration of CNDP fighters into the national army and better political representation. It was formed in 2012 after claims that the agreement was not honored, M23 is composed primarily of ethnic Tutsis, a minority group in eastern DRC. It operates mainly in North Kivu province, near the borders with Rwanda and Uganda**. So, pair 2 is correctly matched.**
- **Majeed Brigade:** It is an offshoot of the Baloch Liberation Army (BLA), the most prominent separatist group operating in **Pakistan's Balochistan province**. Active since 2011, the unit is named after two brothers, Majeed Langove Sr. and Jr., who became symbols of Baloch resistance after sacrificing their lives for the cause of Baloch autonomy and freedom. **So, pair 3 is not correctly matched.**
- **Al-Shabaab:** It is a designated terrorist organization and is considered one of the most dangerous jihadist groups in **Africa**. It emerged in 2006, evolving from the remnants of al-Ittihad al-Islamiya (AIAI), a militant Islamist group that rose after the collapse of **Somalia**'s central government in 1991. It is an affiliate of al-Qaeda and has been responsible for numerous suicide bombings, mass shootings, and terror attacks across **Somalia** and neighboring countries, especially Kenya. **So, pair 5 is not correctly matched.**

So, only two of the above given pairs are correctly matched.
Therefore, option (b) is the correct answer.

Q46.
Answer: d
Explanation:

- The **'75/25 initiative'** launched by the **Ministry of Health and Family Welfare** on **World Hypertension Day (May 17, 2023)**, aims to provide **standardized care to 75 million people** with **hypertension and diabetes** by **December 2025**.
- This initiative falls under the broader **National Programme for Prevention and Control of Non-Communicable Diseases (NP-NCD)**, which focuses on early detection, opportunistic and population-based screening, diagnosis and treatment support, teleconsultation, and awareness promotion.

- Hypertension is a NCD which presents as increased Blood Pressure in the affected individuals and is linked to many systemic ailments such as glaucoma, cardiac myopathies, renal failure etc. It may remain undetected in individuals for years. With changing lifestyles to a sedentary mode the incidence of the disease has increased at a worrisome pace, especially amongst the young individuals.

Therefore, option (d) is the correct answer.

Relevance: The Minister of State for Health and Family Welfare gave a written reply to the Parliament regarding the progress of 75/25 initiative.

Q47.

Answer: b

Explanation:

- **Large Phased Array Radar (LPAR)**, also known as phased array electronic scanning array radar, utilizes a large number of individually controllable small antenna elements arranged in an antenna array.
- These are an **advancement over the rotating antenna technology**, which have limited target capacity, and cannot detect high-speed aircraft and artificial Earth satellites, these mechanical radars are unable to adapt to larger coverage areas, higher data rates, and environments with multiple targets. LPARs, on the other hand, **use a directional electromagnetic wave beam emitted by each individual antenna** in a specific direction and these beams are superimposed to achieve directional scanning. **So, statement 1 is not correct.**
- Unlike conventional radars that rely on mechanical rotation, LPARs use electronically controlled antennas that scan vast areas almost instantly. **This allows them to track multiple objects with precision, making them a crucial component of modern air defence networks**. All ballistic missiles, including those armed with nuclear warheads, travel through low-earth orbit before re-entering the atmosphere. Advanced radar systems like LPAR are specifically designed to track these trajectories. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.

Relevance: China has recently installed a LPAR in its Yunnan province, generating security concerns for India

Q48.

Answer: c

Explanation:

- **Indian Spiny Tailed Lizard (*Saara hadwick*)** is an **endemic reptilian species** of the arid grassland regions of Northern Indian plains. The lizard, found on hard sandy and gravel plains, lives in large colonies or lounges, but lives singly in burrows dug by it. It is part of the diet of other desert fauna like raptors, desert foxes and snakes.
- The reptile is diurnal, spending most of its day basking and foraging, and returning to its burrow at sunset for safety. Interestingly, **it is the only herbivorous lizard species in the country**, feeding on local grasses of the region. **So, statement 1 is correct.**
- It has a distinct **fat tail covered in sharp spines, where it stores fat for its long hibernation season** – an evolutionary strategy to survive in the arid ecosystem. **So, statement 2 is correct.**

- Locally known as '*sandho*' or '*sanda*', this **species is increasingly poached for its fatty tail,** the oil derived from the tail is sold as *'sande ka tel*,' a popular aphrodisiac widely used across South Asia. For the rapid depletion of the species, it is **listed as 'vulnerable'** in the IUCN red list. **So, statement 3 is correct.**

So, all of the above statements are correct.
Therefore, option (c) is the correct answer.

Q49.
Answer: c
Explanation:

- **Red rot**, caused by the fungus *Colletotrichum falcatum*, is a **major disease of sugarcane**, commonly known as *laal saadan* due to the **reddish discoloration of the internal stalk tissue**.
- It spreads through air, water, setts (pieces of sugarcane stalk used to plant new crops) and soil, reducing the yield in the affected farm, lowering the quality of sugarcane juice and reducing production of sugar, jaggery and bagasse.

Therefore, option (c) is the correct answer.
Relevance: An outbreak of red rot in the sugarcane crops has been reported in the Western Uttar Pradesh in the region.

Q50.
Answer: a
Explanation:

- **Indian Sandalwood (*Santalum album*)** is one of India's most **economically valuable tree species**, prized for its fragrant heartwood and essential oil. Native to the Indian subcontinent, it has historically been referred to as the **"Royal Tree".**
- Growing as a tree in the tropical forests of Western Ghats, *Santalum* is a **root parasite** on nearby trees. It establishes connections with the roots of nearby host plants to **derive water and nutrients**, especially in the early stages of growth, though it can photosynthesize on its own. **So, statement 1 is correct.**
- This plant is **native to several parts of peninsular India**, including **Karnataka, Tamil Nadu, and Kerala**, and is cultivated across various southern states, in the **Western Ghats** region. It also naturally occurs in parts of Southeast Asia and Australia. **So, statement 2 is not correct.**
    - It is Red Sanders, locally known as 'laal chandan' which is endemic to tropical dry forests of Eastern Ghats.
- Indian Sandalwood (*Santalum album*) is **classified as "Vulnerable"**,in the **IUCN Red List,**. mainly due to over-exploitation, illegal trade, and habitat loss. **So, Statement 3 is not correct.**
    - Red Sanders is listed as endangered on the IUCN red list.

So, only one of the above statements is correct.
Therefore, option (a) is the correct answer.

Q51.
Answer: d
Explanation:

- **Stockholm International Peace Research Institute (SIPRI)** releases an annual report regarding **international arms trade** since the early 1990s, this report is taken as a marker for countries' threat assessment, diplomatic engagements, economic situation amongst others.

- **India ranked second in global arms imports during the assessment years of 2020–24 with an 8.3% share. Ukraine topped the list** due to the massive surge in imports related to its war with Russia. India's position dropped from first to second as its imports decreased by 9.3% during this period, partly due to growing domestic defense production. **So, statement 1 is not correct.**
- Despite the declining share of Russian arms in India's imports (from 72% in 2010–14 to 36% in 2020–24), **Russia remains India's top arms supplier**. The USA, France, and Israel are gaining importance, but no country has yet overtaken Russia as of 2024. **So, statement 2 is not correct.**

Therefore, option (d) is the correct answer.

Knowledge Box:

• Top 5 Arms Importers (2020–24): Ukraine, India, Qatar, Saudi Arabia, Pakistan
• Top 3 Suppliers to India:
  o Russia (36%)
  o France
  o Israel / USA
• **USA:** World's largest arms exporter (43% of global share), but not India's top supplier
• **Trend:** India is diversifying away from Russian arms due to delays, war constraints, and self-reliance goals

Q52.

Answer: a

Explanation:

- The Kiel Canal is one of the most strategically important artificial waterways in the world, **connecting the North Sea with the Baltic Sea**. It enables ships to bypass the longer and more hazardous route around Denmark's Jutland Peninsula, saving over 500 km. The canal stretches approximately 98.6 kilometers and is entirely located in Germany, between Brunsbüttel on the North Sea and Kiel-Holtenau on the Baltic Sea.

Therefore, option (a) is the correct answer.

Knowledge Box:
• **Length of Canal:** ~98.6 km • **Jurisdiction:** Entirely with Germany (not on a border) • **Connected Ports:** Brunsbüttel (West), Kiel (East) • **Legal Status:** Originally internationalized under the Treaty of Versailles (1919), later brought fully under German control.

Q53.
Answer: d
Explanation:

- Satellite internet or satellite broadband is a wireless internet connection provided through communication satellites orbiting the earth. It is **independent of location and can be accessed from anywhere within the range of satellites,** as it provides global coverage.
- Satellite internet is emerging as an **effective alternative for communication and broadband services in remote and rural areas** where other traditional internet mediums like digital subscriber line (DSL) and cable are hard to reach.
- **Satellite spectrum**, is a shared resource unlike the terrestrial services, which require exclusive use of spectrum and, therefore, limited spectrum usage-efficiencies, **satellite spectrum can be shared by multiple operators,** and represents excellent efficiency of spectrum usage. Importantly, **satellite spectrum is managed by the International Telecommunication Union (ITU)** through a global convention, which is signed by 194 nations, including India. **So, statement 1 is not correct.**
- The basic prerequisite of a resource that is to be auctioned is that it should be available for sale as discrete, unique products. Satellite spectrum does not satisfy this elementary criterion, because the concerned satellite band of frequencies is shared by all operators. **The terrestrial concept of exclusivity does not apply in the case of satellite spectrum** and, therefore, terrestrial-type pricing is not applicable.
- The Telecommunications Act, 2023 takes into cognisance this requirement of the satellite based internet and **provides for the administrative allocation of the spectrum,** instead of the auction based allocation. **The satellite-based services are listed in the First Schedule of the Act**. **So, statement 2 is not correct.**

Therefore, option (d) is the correct answer.
Relevance: Starlink has signed a deal with Bharati Airtel and Jio for provisioning of high speed internet in India.

Q54.
Answer: a
Explanation:

- The **National Pharmaceutical Pricing Authority (NPPA),** under the Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, fixes the **ceiling prices for essential medicines as per the National List of Essential Medicines (NLEM),** which is issued by the Ministry of Health and Family Welfare. This is done under the Drugs (Prices Control) Order (DPCO), 2013, which is guided by the National Pharmaceutical Pricing Policy (NPPP), 2012. **So, statement 1 is correct.**
- The **Drugs and Cosmetics Act, 1940 governs only the drug quality**, safety, efficacy, manufacturing, and distribution. It does not deal with drug pricing. **Drug Price Control Order, 2013** which is the primary statute for **controlling the drug pricing is issued under the Essential Commodities Act, 1955**. **So, statement 2 is not correct.**

- The Central Drugs Standard Control Organization (CDSCO) is the nodal agency for drug approval in India, but **it functions under the Ministry of Health and Family Welfare,** not the Ministry of Chemicals and Fertilizers. It handles approvals, clinical trials, and regulation of medical products. **So, statement 3 is not correct.**

So, only one of the above statements is correct.

Therefore, option (a) is the correct answer.

Relevance: The Minister of State for Chemicals and Fertilizers in an oral reply to the Parliament provided information on the drugs and the mechanisms the prices of drugs being controlled.

Q55.

Answer: d

Explanation:

- **Remittances** involve the transfer of money between individuals, which can take the form of a bill, an invoice or a gift. More specifically, the term 'remittance' **encompasses the funds that migrants send to their family members in their country of origin while they are employed and residing in a foreign nation.**
- As per the recent World Migration Report, **India remains the world's top recipient of remittances, with over $118 billion received in the Financial Year 2023–24.**
- According to the RBI's Sixth Round of the Remittances Survey (2025), **the United States is now the largest source of India's inward remittances, contributing 27.7% in 2023–24.** In contrast, the **United Arab Emirates (UAE)**, which was once the top source, accounted for **19.2%,** making it the second largest source. This shift reflects a change in the migration pattern, from low-skilled labour migration to Gulf countries to high-skilled emigration to advanced economies, especially the US and UK. **So, statement 1 is not correct.**
- The state-wise remittance data from the same RBI report shows that **Maharashtra received the highest share at 20.5%, followed closely by Kerala at 19.7%.** Given the survey's categorization and the way remittances are distributed, Kerala is often recognized as the state most dependent on remittances. However, in absolute terms, Maharashtra received more in 2023–24. **So, statement 2 is not correct.**

Therefore, option (d) is the correct answer.

Knowledge Box
• **Top Source Countries for Remittances to India (2023–24):** o United States: 27.7% o UAE: 19.2% o UK: 10.8% o Singapore: 6.6% o Saudi Arabia: 6.7% • **Top Recipient States (2023–24):** o Maharashtra: 20.5% o Kerala: 19.7% o Tamil Nadu: 10.4%

- Telangana: 8.1%
- Karnataka: 7.7%

- **Key Trend:** There has been a notable shift from Gulf nations to advanced economies as remittance sources, driven by an increase in high-skilled migration.

Relevance: The Reserve Bank of India (RBI) has recently released the sixth round of Remittance Survey (2025).

Q56.
Answer: c
Explanation:

- The **Animal Birth Control (ABC) Rules, 2023,** were introduced by the Central Government to standardize and **regulate the sterilization and immunization of stray animals across India.**
- The ABC Rules are notified under the **Prevention of Cruelty to Animals Act, 1960**. This Act serves as the primary legislation for ensuring the **humane treatment of animals** and preventing cruelty toward them.
- **The ABC Rules, 2023** provides a legal framework for sterilization and vaccination of stray dogs while directing municipal bodies to implement and fund these programs at the same time promoting humane animal control as an alternative to culling or relocation.

Therefore, option (c) is the correct answer.
Relevance: The Government of India in a written reply in the Parliament, mentioned about the steps taken for stray animal welfare and the sterilisation campaigns taken as per the Animal Birth Control rules, 2023.

Q57.
Answer: c
Explanation:

- Fungal diseases, also known as **mycoses**, are infections caused by organisms belonging to Kingdom fungi and affect various parts of the body, including the skin, hair, nails, and even internal organs. Common examples include ringworm, athlete's foot, and yeast infections.
- **Candidiasis** is a fungal infection caused by ***Candida*** species, which often affects the skin, mouth, and genital areas. It can lead to oral thrush, vaginal yeast infections, and systemic infections in immunocompromised individuals. **So, point 1 is correct.**
- **Syphilis** is a sexually transmitted disease caused by the **spirochete bacterium *Treponema pallidum***. It progresses through several stages, causing sores, rashes, and in severe cases, damage to organs like the heart and brain. **So, point 2 is not correct.**
- **Ringworm** is a **fungal infection,** which affects the skin, hair, and nails. It forms a ring-shaped rash, often on the scalp, feet, or body, and is highly contagious. **So, point 3 is correct.**
- **Athlete's Foot** (tinea pedis) is a fungal infection caused by dermatophytes, particularly ***Trichophyton*** species. It affects the feet, causing itching, redness, and peeling skin, especially between the toes. **So, point 4 is correct.**
- **Mucormycosis**, also known as "**black fungus,**" is a rare but serious systemic fungal infection caused by fungi in the **order Mucorales.** It typically affects people with weakened immune systems and can lead to severe complications if untreated. **So, point 5 is correct.**

So, only four of the above given diseases are caused by fungus.

Therefore, option (c) is the correct answer.

Knowledge Box
● **MycoNet Program by ICMR:** o Launched in 2020, MycoNet maps fungal infections across India, focusing on invasive fungal infections (IFIs). It includes eight Advanced Mycology Diagnostic and Research Centers (AMDRCs) that collaborate with state governments and medical colleges. ● **AMR Surveillance:** o ICMR's Antimicrobial Resistance Research and Surveillance Network (AMRSN) monitors antifungal resistance in key pathogens like Candida, Cryptococcus, and Aspergillus. The National Centre for Disease Control (NCDC) integrates this surveillance into the National AMR Surveillance Network (NARS-Net).0

Relevance: In a written reply to the Parliament, the Government of India highlighted the steps taken to prioritize research of Fungal infections.

Q58.
Answer: a
Explanation:

- The **Bharatiya Nyaya Sanhita (BNS), 2023** which replaced the archaic Indian Penal Code (IPC), 1871, have included a separate chapter (**Chapter V**) for '**offences against women and child**', earlier the IPC covered these offences under Chapter XVI which was covered the 'offences against human body'. **So, statement 1 is correct.**
- While life imprisonment is provided for many offences (e.g. gangrape of a woman below 18 years), the **maximum punishment under BNS is actually death sentence** for certain grievous crimes. **So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Knowledge Box
Provisions for protection of Women and Children under BNS: ● Offences against woman and child have been given precedence over all other offences in a new Chapter-V of BNS. ● Various offences against women and children have been made gender-neutral in BNS, covering all victims and perpetrators regardless of gender. ● In BNS, the age differential for minor victims of gangrape has been done away with. Earlier different punishments were prescribed for gangrape on a girl below the age of 16 years and 12 years. This provision has been modified and now gangrape on a woman below the age of eighteen years is punishable with imprisonment of life or death.

Relevance: The Minister of State for Home Affairs in a reply to the Parliament underscored about the provisions of Bharatiya Nyaya Sanhita regarding offences against women and Child.

Q59.
Answer: c
Explanation:

- The Ministry of Home Affairs, under its **Indian Cyber Crime Coordination Centre (I4C)** initiative, has launched the **Samanvaya** and **Sahyog** platforms to strengthen the prevention and investigation of cybercrimes.
- **Samanvaya** acts as a Management Information System (MIS) platform for law enforcement agencies. It enables **inter-state data sharing and analytics of cybercrime-related information**, and helps establish linkages between crimes and criminals. The *Pratibimb* module of this platform maps the locations of cybercriminals and related infrastructure, facilitating faster investigation.

Therefore, option (c) is the correct answer.

Knowledge Box
• The Indian Cyber Crime Coordination Centre (I4C) is an initiative under the Ministry of Home Affairs in India, launched in January 2020 to address the growing concerns of cybercrime within the country. • **Objective**: The objective of I4C is to create a centralized system for the detection, prevention, and investigation of cybercrimes, which have been on the rise with increased digitization.

Q60.

Answer: a

Explanation:

- The PM-DAKSH (Pradhan Mantri Dakshta Aur Kushalta Sampann Hitgrahi) scheme is a **Central Sector Scheme aimed at improving the employability of socially and economically marginalized groups**. It is implemented directly by the central government without involving state-level funding or implementation roles. The scheme focuses on both wage and self-employment through skill development training. **So, statement 1 is not correct.**
- The scheme targets **Scheduled Castes (SCs), Other Backward Classes (OBCs), Economically Weaker Sections (EWSs), Denotified Tribes (DNTs) and Safai Karamcharis including waste pickers for skill training**. This wide coverage ensures inclusive skill development. **So, statement 2 is not correct.**
- Under this scheme, **any OBC and EWS candidates whose annual family income is less than Rupees 3 lakh are eligible to get training** whereas there is no income limit for candidates belonging to SC/ DNT/ Safai Karamcharis (Safai mitras) including waste pickers. **So, statement 3 is correct.**

So, only one of the above given statements is correct.

Therefore, option (a) is the correct answer.

Q61.
Answer: c
Explanation:

- **Sujni embroidery**: It is a GI tagged traditional textile craft from **Bihar, India,** known for its vibrant, intricate designs and storytelling through needlework. It involves layering fabric, often old saris or dhotis, and then embroidering designs onto the top layer using a combination of running and chain stitches. **So, pair 1 is not correctly matched.**
- **Kantha embroidery**: It derives its name from *kontha*, Sanskrit word for rags. This art involves crafting fabrics using patchwork from old and worn out clothes and is currently practiced in the States of **West Bengal**, Odisha and adjoining regions in Bangladesh. **So, pair 2 is correctly matched.**
- **Kheta embroidery:** It is a traditional and lesser-known art form practiced by the Shershabadi community in **Bihar**. The embroidery is deeply rooted in local culture and nature, particularly the life around rivers, and has been recently brought into the spotlight through exhibitions and expert-led documentation. **So, pair 3 is correctly matched.**
- **Kasuti embroidery:** It is a counting thread embroidery in which the stitches of the design are measured in terms of the number of base fabric threads in the area to be embroidered. The State of **Karnataka** has been granted GI tag for this craft. **So, pair 4 is correctly matched.**

So, only three of the above given pairs are correctly matched.
Therefore, option (c) is correctly matched.
Relevance: The Government in the latest session of Parliament, submitted the steps taken by it for the progress of Kheta embroidery on the floor of the house.

Q62.
Answer: c
Explanation:

- Saffron, derived from the *Crocus sativus* flower, is one of the most prized spices globally due to its flavour, colour, aroma, and medicinal value. **It is believed to have entered India through ancient trade routes like the Silk Route**, and it continues to be grown primarily in Jammu & Kashmir in India.
- What makes saffron particularly expensive is not the difficulty in growing it, but the **labor-intensive process involved in its harvesting**. Each flower yields only three stigmas, which must be handpicked in the early morning hours when the flower is fully open. This process requires tens of thousands of flowers to produce just one kilogram of saffron, demanding hundreds of hours of work.
- Another unique feature of **saffron is that it does not propagate via seeds.** The ***Crocus sativus* plant is sterile**, so farmers must dig up and replant the corms (underground bulb-like structures) each season. It thrives in semi-arid regions with well-drained soil but is highly sensitive to temperature extremes, making its cultivation geography-specific.
- The **topmost producer of the spice is Iran** followed by India, where it is grown exclusively in Kashmir.

Therefore, option (c) is the correct answer.
Relevance: The Government is pushing for extending cultivation of Saffron to the state of Sikkim as was done for the case of Lavender.

Q63.
Answer: b
Explanation:

- Hyperloop is a proposed mode of ground transport for passenger and freight transportation which consists of **pods transported at high speed**, through the length of low-pressure tubes that are elevated off the ground. Hyperloop has the potential to travel at speeds in excess of 1125km/h which would reduce travel times over medium-range distances.
- **Hyper loop is based on a principle of magnetic levitation.** The principle of magnetic levitation states that the vehicle can be suspended and propelled on a guidance track made with magnets. The capsule on the top of the track is propelled by linear induction motors. That is the **vehicle and the track have no physical contact, obliterating the scope for frictional force,** thus limiting resistance caused by frictional force to the fast moving vehicle. **So, statement 1 is not correct.**
- The **capsules are accelerating at a high speed due to linear electric motors and move above their track with the help of passive magnetic levitation or air bearings.** Powerful electromagnets levitate and propel the capsules forward. This reduces the frictional losses, as trains glide over the desired track. As there are no frictional losses, the capsules travel at a very high speed.
- This technology **requires the creation of vacuum tubes.** The low pressure air in the tube also **increases the efficiency by dismissing the losses due to aerodynamic drag** and this makes the capsule travel at very high speed. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.
Relevance: The Union Railway Minister has announced India's first prototype of hyperloop technology to be built at IIT Madras.

Q64.
Answer: d
Explanation:

- India regularly conducts bilateral and multilateral naval exercises to enhance interoperability, maritime security, and defense cooperation. These exercises serve strategic, operational, and diplomatic purposes by engaging with key partner nations across regions.
- **VARUNA:** It is a bilateral maritime drill conducted between the **Indian Navy and the French Navy**. It has been held annually since 2001 and includes high-end warfighting training such as anti-submarine operations, air defense maneuvers, and coordinated fleet actions. This exercise symbolizes the growing Indo-French strategic partnership in the Indo-Pacific region. **So, pair 1 is correctly matched.**
- **BONGOSAGAR:** It is a bilateral exercise between the **Indian Navy and Bangladesh Navy**, aimed at enhancing interoperability and cooperation in maritime operations. Initiated in recent years, the exercise includes naval drills, seamanship activities, and coordinated patrols (CORPAT). It reflects the strong defence cooperation between the two neighbouring countries. **So, pair 2 is correctly matched.**
- **INDRA**: It is a long-standing joint military exercise between **India and Russia**, which includes components from the Army, Navy, and Air Force. The naval component of Indra features complex maritime operations such as anti-piracy, boarding actions, and tactical navigation. It exemplifies the strategic military partnership and trust between India and Russia. **So, pair 3 is correctly matched.**

- **Naseem-Al-Bahar**: Literally meaning "Sea Breeze", is a **bilateral naval exercise between India and Oman.** It involves surface warfare, communication procedures, and search and rescue operations. Conducted since 1993, this exercise enhances maritime security collaboration and operational coordination with a key Gulf partner. **So, pair 4 is correctly matched.**

So, all four of the above given pairs are correctly matched.
Therefore, option (d) is the correct answer.

Q65.
Answer: d
Explanation:

- The **Governor's address** in the State Legislature is a **ceremonial function** mandated by the **Constitution of India** under **Article 176**. It takes place at the beginning of the **first session after each general election** and the **first session of each year**. This address outlines the **policies and agenda** of the **State government** as advised by the **Council of Ministers**, which the Governor must read out **without alteration**.
- **Article 176** of the **Indian Constitution** mandates the Governor to **address the State Legislature** at the beginning of each legislative session. This is a **constitutional responsibility**, and the address is used to inform the legislature about the **government's agenda** for the upcoming session. **So, statement 2 is correct.**
- During the Constitutional Assembly Debates (CAD), the issue for providing specificity to the President and Governor's address was made, but that remained rather unsuccessful, leaving the **Constitution silent on the matters of the adherence of the President and State Governor's adherence to the text of the speech** as provided by the Cabinet or the Council.
- The courts in the *Andur Gafoor Habibulah versus the Speaker of West Bengal* legislative assembly said that **Governor cannot decline to deliver his/her address at it is a Constitutional duty of the office,** but if the Governor chose to walk out, leaving the speech on the table, it would be **considered an irregularity not an illegality**. In a further case, the court stated that the **Governor can exercise his/her discretion in not delivering and leaving out irrelevant sections of the speech**. **So, statement 1 is not correct.**

So, statement-I is incorrect, but Statement-II is correct
Therefore, option (d) is the correct answer.
Relevance: Recently, during the Budget address of the state legislative assembly, the Gobernor of WB deviated from his speech.

Q66.
Answer: d
Explanation:

- As per the recent Economic Survey, systematic deregulation is as critical, if not more, as investments in infrastructure and incentives to encourage innovation and create a viable business atmosphere. According to the Survey, the effect of excessive regulation is more profound on Small and Medium enterprises (SMEs), while the large enterprises have enough coffers to pay for the obligation of regulations, these SMEs often chose to remain small to avoid regulation.
- The theme of deregulation in the recent years have become the central theme for economic and political policy making.:

- The **United States of America has recently established a new Department of Government Efficiency.** This department has been set up with an objective of streamlining government functioning, eliminating waste, and upgrading critical infrastructure, at the same time downsizing the Government controls on business. **So, pair 1 is correctly matched.**
- **The United Kingdom Parliament has adopted the Better Regulation Framework (BRF)** to manage the flow of regulation and understand its impacts. As part of this framework, the Parliament enacts all new legislation in adherence to the '**one-in, two-out**' principle. This principle requires every pound of new regulatory burden on businesses to be offset by identifying and removing two pounds of existing burdens from other areas of the regulation. **So, pair 2 is correctly matched.**
- **New Zealand has recently established a Ministry of Regulation.** It has been established to examine laws and propose repeals and amendments to improve compliance experience. It is aimed at lifting quality across all regulatory systems, and supporting agencies with regulatory responsibilities to align with good practices. **So, pair 3 is correctly matched.**

Therefore, option (d) is the correct answer.

Q67.
Answer: a
Explanation:

- **Market correction** is taken to as a drop of 10% or more in stocks from their most recent high. The 10% trigger for a correction is an arbitrary, round-number threshold. But it serves as a signal that investors have turned pointedly more pessimistic about the market. Economic Survey 2024-25 cautions against 'meaningful' market correction in 2025 due to elevated valuations and optimistic market sentiments in the U.S.
- The S&P 500 of U.S has experienced a significant decline, falling 10.1% since its February 19 peak due to concerns over President Donald Trump's tariffs from its recent highs. A downturn in the US market could have **a ripple effect in India, by significantly affecting the investor sentiments and spending habits. So, statement 1 is correct.**
- **This is particularly concerning given the surge in young investors entering the market after COVID.** It is because new investors who joined the market after the **pandemic haven't experienced a major market downturn**. If one happens now, it could **significantly affect investor attitudes and spending habits. So, statement 2 is correct.**

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I
Therefore, option (a) is the correct answer.
Relevance: The aspect that Indian markets may face uncertainty in a scenario of market correction was underscored by the CEA in one of his recent speech while reiterating the provisions of economic survey.

Q68.
Answer: c
Explanation:

- The North Eastern Handicrafts and Handlooms Development Corporation Ltd. (NEHHDC) under the **Ministry of Development of North Eastern Region (MDoNER)** has obtained **Oeko-Tex certification** for **eri silk** from **Germany**.
- **Oeko-Tex** is a **global certification system** that tests **raw, semi-finished, and finished textile products** (like yarns, fabrics, buttons, and linens) for **harmful substances**.
- It ensures that certified products are **safe for human use** and comply with **international safety and environmental standards**.

- The recent certification of **eri silk** under Oeko-Tex boosts its **marketability** in **eco-conscious global markets**.

Therefore, option (c) is the correct answer.

Knowledge Box
Key Facts About Eri Silk: ● **Eri silk** is known as the **world's only vegan silk** because the **moth is not killed** during extraction, making it a **cruelty-free** and **sustainable** textile. ● It is derived from the **domesticated silkworm** species ***Philosamia ricini***, which **feeds on castor leaves**. ● In India, **eri silk cultivation** is mainly practiced in the **north-eastern states**, especially **Assam**, and also in **Bihar, West Bengal**, and **Odisha**. ● **Eri silk from Assam** holds a **Geographical Indication (GI) tag**. ● Government initiatives include: ○ Central Muga & Eri Research Institute at Lahdoigarh, Assam for R&D, training, and extension. ○ R&D by the Central Silk Board to develop improved practices and increase productivity. ○ Muga Eri Silkworm Seed Organization for maintaining and supplying quality eri silkworm seed. ○ Implementation of the Silk Samagra-2 scheme (2021-22 to 2025-26) for holistic development of sericulture, including eri silk.

Relevance: Recently, NEHHDC under the Ministry of Development of North Eastern Region obtained Oeko-Tex certification for eri silk.

Q69.

Answer: c

Explanation:

- India and the United State of America signed the **first-ever Cultural Property Agreement (CPA)** on **26 July 2024** during the **46th World Heritage Committee** at **Bharat Mandapam** in New Delhi.
- The **Cultural Property Agreement (CPA)** between **India and the United States** aims to **prevent illicit trafficking of Indian antiquities and facilitate their repatriation**. As of 2024, 588 antiquities have been repatriated, including 297 in 2024 alone.
- The **CPA restricts the importation into the United States of certain archaeological material** ranging in date from 1.7 million years ago through 1770 CE and certain ethnological material, which may include categories of civic, religious, and royal architectural material, religious material and ceremonial items, and manuscripts ranging in date from 2nd century BCE to 1947 CE. **So**, **statement 1 is correct.**
- The agreement is **aligned with the 1970 UNESCO Convention** on the Means of Prohibiting and Preventing the Illicit Import, Export and Transfer of Ownership of Cultural Property, to which both countries are States party. It provides for **technical cooperation, mutual understanding**, and the return of **designated cultural materials** ranging from **prehistoric to 1947 CE**. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.

Knowledge Box

Key Facts About Cultural Property Agreement (CPA):

- The CPA is **preventive in nature**, without timelines or numerical targets.
- It supports India's collaboration with **UNESCO and INTERPOL** to combat cultural property crimes.
- The CPA allows **quick seizure and repatriation** of Indian antiquities intercepted by **US Customs**.
- The agreement was a result of **Article 9 of the 1970 UNESCO Convention**, bilateral negotiations, and a **Statement of Fact** documenting cultural cooperation needs.
- The **'Antiquities Coalition'**, a US-based NGO, played a key role in supporting the agreement.
- The **Kashi Culture Pathway**, part of India's G20 presidency, emphasized the **protection of cultural property** and influenced the finalization of the CPA.

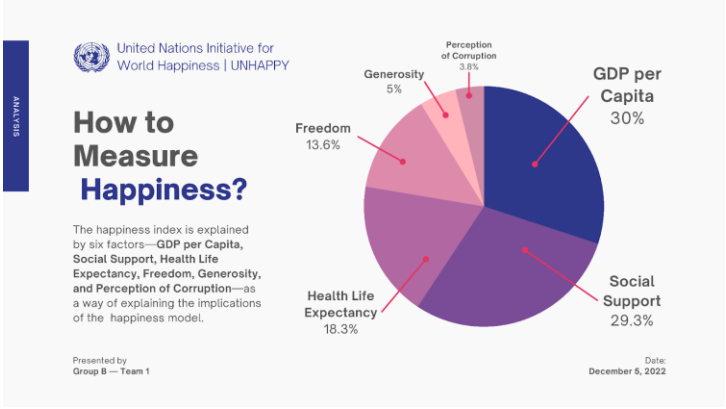
Relevance: The Government of India and the United States of America signed the first ever 'Cultural Property Agreement' to prevent and curb the illicit trafficking of antiquities from India.

Q70.

Answer: a

Explanation:

- The Wellbeing Research Centre at the **University of Oxford** in partnership with **Gallup**, the **UN Sustainable Development Solutions Network (UNSDSN**) has published the **World Happiness Report** (WHR) 2025 on World Happiness Day (20th March).
- The **World Happiness Index** is primarily based on responses to a **single life evaluation question** called the **Cantril Ladder**, where people rank their current life on a scale of 0 to 10. This **life evaluation** serves as the core metric and is collected via the **Gallup World Poll**.



- While the ranking is not a simple index of multiple metrics, the **life evaluation scores are statistically explained** using several **predictor variables**, which include:
  - **GDP per capita** (economic prosperity)
  - **Healthy life expectancy** (health and longevity)
  - **Social support** (reliance on others in times of need)

- **Freedom to make life choices**
- **Perceptions of generosity and corruption**

- **Access to the internet, emotions and mental states** are **not directly used in ranking**, as they fluctuate day-to-day and are considered less stable.

Therefore, option (a) is the correct answer.

Knowledge Box
Key Facts About World Happiness Index 2025: ● **India ranks 118th** out of 147 countries in the 2025 report, far below the top 50. ● **Finland** has maintained its position as the **world's happiest country for the 8th consecutive year**, with a high happiness score of **7.736**. ● Both the **United States (24th)** and the **United Kingdom (23rd)** saw declines in their happiness rankings, attributed to **rising social isolation, political polarisation**, and **economic uncertainty**. ● **Costa Rica** and **Mexico** are notable **new entries** into the top 10, emphasizing that **happiness is not solely tied to GDP**, but also **social cohesion and cultural values**. ● **Afghanistan** remains the **unhappiest country**, followed by **Sierra Leone** and **Lebanon**.

Relevance: The World Happiness Report 2025 highlights Finland as the happiest country for the 8th time.

Q71.
Answer: c
Explanation:

- **Navroz** is the **Persian New Year festival** celebrated. It is observed with great enthusiasm by the **Parsi community** in India and other Zoroastrian communities worldwide.
- **Navroz**, also known as **Nowruz**, is celebrated on the **vernal equinox** (around March 21), symbolizing the **beginning of spring** and the **Persian New Year**. It holds seasonal and cultural significance among **Zoroastrians and Parsis**. **So**, **statement 1 is correct.**
- Navroz has been **recognized by UNESCO** and is listed in the **Intangible Cultural Heritage of Humanity**, acknowledging its **rich cultural traditions** and **global significance**. **So**, **statement 1 is correct.**

Therefore, option (c) is the correct answer.

Knowledge Box
Key Facts About Navroz: ● **Navroz** (also spelt **Nowruz**) is the **Iranian and Persian New Year**, meaning **"new day"**. ● It marks the **beginning of spring** and is celebrated with enthusiasm by the **Parsi community** across the world. ● The festival is associated with the **Persian King Jamshed**, who introduced the **Shahenshahi calendar**. ● **Navroz in India** is observed around **August 16–17**, as per the **Shahenshahi calendar**, which **does not include leap years**. ● The festival was brought to India by **Nusservanji Kohyaji**, an **18th-century Surat-based Parsi tradesman**.

Relevance: The Prime Minister, Narendra Modi, has greeted everyone on the occasion of Navroz.

Q72.
Answer: d
Explanation:

- **The Armed Forces (Special Powers) Act (AFSPA)** was passed by Parliament in 1958 to grant **special powers** to the armed forces for maintaining **public order** in **"disturbed areas."** It is enforced **only after** an area is declared "**disturbed**" under **Section 2** of the Act.
- As of **April 2025**, the **AFSPA** has been **extended for six months** in the following states:
  - **Manipur**: AFSPA is in force in **most parts of the state**, except in the jurisdiction of **13 police stations** across **five valley districts**. Hence, **AFSPA is partially applicable** in Manipur. **So, point 1 is correct.**
  - **Nagaland**: AFSPA is in force in **eight districts** and **21 police stations** in **five other districts**, meaning it remains operational in a substantial part of the state. **So, point 2 is correct.**
  - **Arunachal Pradesh**: AFSPA is applicable in **Tirap, Changlang, and Longding districts**, and parts of **Namsai district** (in police stations of Namsai, Mahadevpur, and Chowkham). **So, point 3 is correct.**
  - **Assam: Only three of Assam's 35 districts-Tinsukia, Sivasagar and Chariadeo districts**, now remain as "disturbed areas" under the AFSPA as the banned ULFA-I has its bases in neighbouring Mynamar and is active in these three districts. **So, point 4 is correct.**

So, all four of the above states have the AFSPA in force as of April 2025.
Therefore, option (d) is the correct answer.

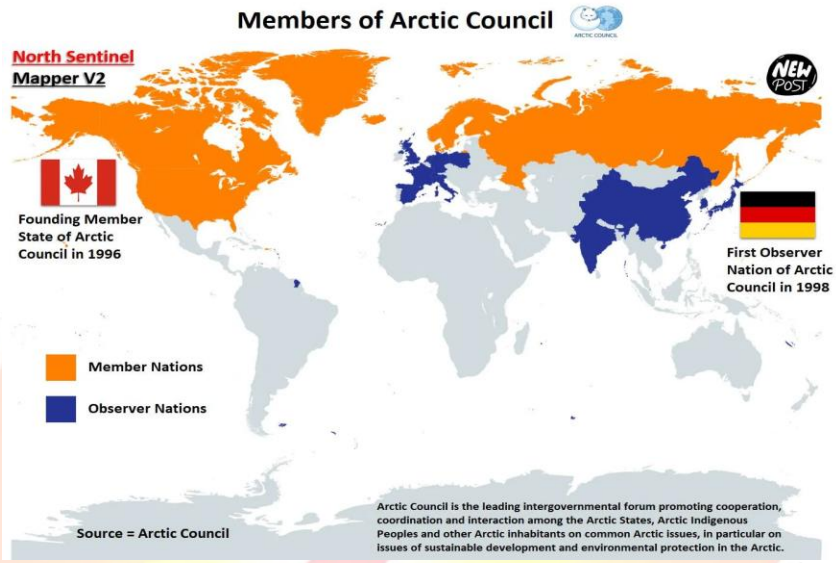
Knowledge Box

Key Facts About Armed Forces (Special Powers) Act (AFSPA):
● The **Central Government**, the **Governor of a State**, or the **Administrator of a Union Territory** can declare any region as a disturbed area under **Section 3** of the Act.
● Areas can be declared disturbed due to conflicts based on **religion, race, language, region, or caste**.
● Once declared, a disturbed area remains so for a **minimum of three months**, as per the **Disturbed Areas (Special Courts) Act, 1976**.
● In **Nagaland and Arunachal Pradesh**, periodic notifications are issued by the **Union Ministry of Home Affairs (MHA).**
● **Special powers granted**:
  ○ **Prohibit public gatherings** of five or more persons.
  ○ Use **force or open fire** after warning, if law is contravened.
  ○ **Arrest without warrant**, **search premises without warrant**, **ban firearms**.
  ○ **Hand over arrested persons to the nearest police station** with a report.
  ○ **Legal immunity**: Armed forces **cannot be prosecuted** without **prior sanction** from the **Union Government**.

Relevance: The Union Ministry of Home Affairs (MHA) extended the AFSPA, 1958 for another six months in the states of Manipur, Nagaland, and parts of Arunachal Pradesh.

Q73.
Answer: b
Explanation:

- The **Arctic Council** is an **intergovernmental forum** established in **1996** through the **Ottawa Declaration** for promoting **cooperation** among **Arctic States and Indigenous communities** on **environmental protection** and **sustainable development**.
- The **Ottawa Declaration**, signed in **1996**, led to the **establishment of the Arctic Council**, an intergovernmental forum promoting cooperation among Arctic states and Indigenous communities on issues such as sustainable development, environmental protection, and scientific research.
- India has been an **observer to the Arctic Council since 2013**.



Therefore, option (b) is the correct answer.

Knowledge Box

About Arctic Council:
• **The Arctic Council** is governed mainly by **UNCLOS (United Nations Convention on the Law of the Sea)**, unlike **Antarctica**, which is protected by **international treaties**. This allows **territorial claims**, **resource exploitation**, and **military presence** in the Arctic.
• **Member States (8)**: **Canada**, **Denmark (via Greenland)**, **Finland**, **Iceland**, **Norway**, **Russia**, **Sweden**, and the **United States**.
• These countries control **land territory** in the Arctic and have rights over resources in their **Exclusive Economic Zones (EEZs)**.
• **Permanent Participants**: **Six Indigenous groups** that represent Arctic inhabitants. Their consultation is required for all major decisions.
• **Observers**: **13 countries** (e.g., **India**, **China**, **Japan**, **UK**, **France**)
  ○ **13 intergovernmental organizations**
  ○ India participates in Arctic Council activities and conducts research via the Himadri station in Svalbard, Norway.
• **Consensus-based decision-making**: All eight Arctic States must agree, with consultation from permanent participants.

Relevance: Recently, India's role as a stabilising force in the Arctic and its observer status in the Arctic Council have come into focus.

Q74.
Answer: a
Explanation:

- **Spoofing** is a form of **cyberattack** where **false GPS signals** are transmitted to **mislead navigation systems**, affecting both aircraft and drones. This is a **serious threat** in regions experiencing **military activity**, and is increasingly used to **disrupt drone operations** and **commercial aviation**.
    - An example includes the **Azerbaijan Airlines crash** (December 25, 2024), where GPS interference was linked to **Russian air defence** activities against Ukrainian drones.
- GPS spoofing is a **cyberattack technique** where **false GPS signals** are sent to misguide navigation systems about their actual location. It has been increasingly reported in **conflict zones** and along sensitive borders, including the **India–Pakistan and India–Myanmar** regions. **So**, **statement 1 is correct.**
- Due to the **inherently weak signal strength of GPS satellites** when they reach the Earth's surface, they are **particularly susceptible to interference and manipulation**. Spoofing devices **exploit this vulnerability by transmitting stronger, false signals** that can overpower genuine ones, effectively **deceiving navigation systems. So, statement 2 is correct.**
    - The **Azerbaijan Airlines crash** on December 25, 2024, which resulted in 38 fatalities, **was linked to GPS interference**.

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I.
Therefore, option (a) is the correct answer.
Relevance: Instances of GPS interference on passenger aircraft, including 'spoofing' with false signals, are on the rise over conflict zones globally.

Q75.
Answer: c
Explanation:

- **Project PARI (Public Art of India)** is an **initiative of the Ministry of Culture**, Government of India, implemented by the **Lalit Kala Akademi** and the National Gallery of Modern Art. The project aims to promote **public art** that is rooted in **India's millennia-old artistic traditions** (**lok kala/lok sanskriti**) while also embracing **modern themes and techniques**. **So, pair 1 is not correctly matched.**
- The **Sahyog Portal**, developed under the **Information and Technology Act, 2000**, aims to create a **safe and clean cyberspace** for Indian citizens. It enables the **automated issuance of notices** to intermediaries for the **removal or disabling** of access to **unlawful online content**. The Portal brings together authorized government agencies and intermediaries on a single platform to ensure quick and coordinated action against cyber threats. **So, pair 2 is correctly matched.**
- The **BHASKAR Initiative**, launched by the **Ministry of Commerce and Industry** under the **Startup India program**, aims to create a **centralized digital platform** to streamline and enhance collaboration within India's **startup ecosystem**. Its core objective is to build the world's largest digital registry of **startup ecosystem stakeholders**, enabling better **networking**, **resource sharing**, and **visibility**. Startups can qualify for recognition under this initiative if they are **private limited companies, partnership firms, or LLPs** with a turnover of **less than ₹100 crore** in any financial year and are **less than 10 years old**. **So, pair 3 is correctly matched.**

- **SPICED** stands for Sustainability in the spice sector through Progressive, Innovative, and Collaborative Interventions for Export Development. It was launched during the **15th Finance Commission period (2023–24 to 2025–26)**, the scheme aims to **expand cardamom cultivation**, enhance productivity (both small and large cardamom), promote exports, and build institutional and farmer capacity. **So, pair 4 is correctly matched.**

So, only three of the above pairs are correctly matched.
Therefore, option (c) is the correct answer.

Q76.
Answer: c
Explanation:

- Under **Article 222** of the Constitution, the **President**, in consultation with the **Chief Justice of India (CJI)**, can transfer a **High Court judge** from one court to another.
- **As per Article 222(1) of the Constitution,** the **President** may transfer a judge from one High Court to another **after consulting the Chief Justice of India**. **So**, **statement 1 is correct**.
- As per **Article 222(2)**, a **transferred judge** is entitled to a **compensatory allowance** in addition to salary, as may be **determined by Parliament**, and **until so determined**, as **fixed by the President by order**. **So**, **statement 2 is correct**.

Therefore, option (c) is the correct answer.

Knowledge Box
Key Facts About Transfer of a High Court Judge: ● In the **First Judges Case (1981)**, the Supreme Court upheld the **executive's primacy** in transfers. ● However, the **Second Judges Case (1993)** established the **collegium system**, granting primacy to the **CJI** and requiring that transfers serve **public interest** and improve **judicial administration**. ● The **Third Judges Case (1998)** further mandated consultation with the **four seniormost judges** and inputs from Supreme Court judges familiar with the concerned High Court. ● The **transfer process** begins with a collegium recommendation, followed by review by the **Law Minister**, approval by the **Prime Minister**, and final sanction by the **President** through a **gazette notification**.

Relevance: The Supreme Court Collegium on March 24, 2025 recommended the Government to transfer Delhi High Court judge, Justice Yashwant Varma, to the Allahabad High Court.

Q77.
Answer: d
Explanation:

- The **National Institutional Ranking Framework (NIRF)**, launched in **2015 by the Ministry of Education**, is India's official mechanism to rank higher educational institutions. It evaluates institutions based on five broad parameters including **teaching quality, research output, graduation outcomes, inclusivity, and perception**.

- The **National Institutional Ranking Framework (NIRF)** is **not released biannually**; it is an **annual ranking** exercise conducted by the **Ministry of Education** since 2015. **So, statement 1 is not correct**.
- **AQAR** (Annual Quality Assurance Reports) submitted to National Assessment and Accreditation Council **(NAAC)** are **verified by expert committees**, unlike **NIRF** submissions, which are **self-reported by institutions without formal verification**, as highlighted in the recent Madras High Court PIL. **So, statement 2 is not correct.**
- NIRF **does not prohibit** the use of bibliometric data. In fact, it **actively uses data from commercial databases** like Scopus and Web of Science to assess research output under the **Research and Professional Practice (RP)** parameter. **So, statement 3 is not correct**.

So, none of the above statements are correct.
Therefore, option (d) is the correct answer.

Knowledge Box
Key Facts About National Institutional Ranking Framework (NIRF): ● The **NIRF** is a government-adopted **ranking methodology** for **higher education institutions** in India, launched in **November 2015**. ● It evaluates institutions on **objective parameters** and releases **annual rankings** across multiple categories. ● Institutions are ranked based on the following **five parameters**: ○ Teaching, Learning and Resources (TLR) – 30% ○ Research and Professional Practice (RP) – 30% ○ Graduation Outcomes (GO) – 20% ○ Outreach and Inclusivity (OI) – 10% ○ Perception (PR) – 10% ● **Bibliometrics** are **quantitative measures** of research output, including: ○ Number of published papers ○ Citation counts ○ Impact factors of journals ○ Metrics derived from databases like Scopus and Web of Science

Relevance: The Union Minister for Education recently released the India Rankings 2024, prepared under the National Institutional Ranking Framework (NIRF).

Q78.
Answer: c
Explanation:

- The **Collegium System** in India refers to the process by which judges are appointed and transferred in the **Supreme Court and High Courts**.
- As per the **Second Judges Case (1993)** and **Third Judges Case (1998)**, the **Collegium's reiterated recommendations** are **binding on the government**, meaning that if the Collegium resends the same names, the government **must appoint** them. This has been upheld by **Constitution Bench judgments**. **So, statement 1 is correct.**

- To address concerns about **lack of transparency**, the **Supreme Court of India began uploading Collegium resolutions** and **minutes of meetings** on its official website, especially since 2017, following widespread criticism and judicial directions for more transparency. **So**, **statement 2 is correct.**

Therefore, option (c) is the correct answer.

Relevance: An international body of jurists had urged the government and the Parliament to enact a law constituting a 'Judicial Council' to spearhead judicial appointments and transfers.

Q79.

Answer: c

Explanation:

- **The Challenger 150 initiative** is a global scientific collaboration focused on advancing **knowledge of the deep ocean.**
- Endorsed by the United Nations Educational, Scientific and Cultural Organization (UNESCO) as part of the **United Nations Decade of Ocean Science for Sustainable Development (2021–2030),** the initiative promotes inclusive research, capacity building, and standardized data collection.
- It is **named after the historic HMS Challenger** expedition, which laid the foundation for modern oceanography.

Therefore, option (c) is the correct answer.

Relevance: Scientists discovered thriving ecosystems with potential new species beneath the Antarctic ice shelf after the A-84 iceberg broke away from the George VI Ice Shelf.

Q80.

Answer: a

Explanation:

- The **National Policy on Biofuels – 2018 (amended in 2022)** aims to promote sustainable energy alternatives by advancing the target of **ethanol blending in petrol by Ethanol Supply Year (ESY) 2025-26**.
- The **Roadmap for Ethanol Blending in India (2020–25)** notes that the use of **20% ethanol-blended petrol (E20)** causes a **marginal reduction in fuel efficiency** in four-wheelers **designed for E10**, though the impact can be minimized with appropriate **engine tuning**. **So**, **statement 1 is correct**.
- The **National Policy on Biofuels 2018** (amended in 2022) allows **ethanol to be made from food grains** when there is a **declared surplus**, as decided by the **National Biofuel Coordination Committee**. **So**, **statement 2 is correct**.
- Ethanol for blending under the Ethanol Blended Petrol (EBP) Programme is not fully exempt from Goods and Services Tax (GST)**. However,** it is subject to a **reduced GST rate of 5%**. **So, statement 3 is not correct**.
- The Government has clarified that no decision has yet been taken to increase **ethanol blending** beyond **20%**. The current target remains at **20% ethanol blending** by **Ethanol Supply Year (ESY) 2025–26**. **So**, **statement 4 is not correct**.

Therefore, option (a) is the correct answer.

Relevance: The government has advanced the target of 20% ethanol blending in petrol to ESY 2025-26 and achieved 17.98% blending by February 2025.

Q81.
Answer: b
Explanation:

- **Hooch, or spurious liquor,** is typically produced in unregulated, rudimentary conditions, often without proper distillation equipment. The main danger comes from contamination with **toxic chemicals**, especially **methanol**, which is **not adequately separated from ethanol** during the crude distillation process.
- Methanol is responsible for **most fatalities and long-term damage** associated with hooch consumption. It is a highly toxic alcohol that can accumulate in the final product when **temperature control during distillation is absent.** It has a lower boiling point than ethanol and is therefore distilled earlier in the process.
- When ingested, methanol is **metabolized in the human body into formaldehyde and formic acid**, leading to symptoms such as **vision loss, nervous system damage, and, in severe cases, death.**
- Methanol poisoning requires immediate medical treatment. The most widely used **antidote is fomepizole, but due to its high cost and limited availability, ethanol is commonly administered** instead.

Therefore, option (b) is the correct answer.
Relevance: Five policemen given compulsory retirement for involvement in Marakkanam hooch tragedy.

Q82.
Answer: a
Explanation:

- **Anti-dumping duties** are a form of trade remedy used by countries to protect domestic industries from unfair competition. When foreign producers export goods at prices significantly below their normal value (a practice known as "dumping"), it can harm local producers. The **World Trade Organization (WTO) permits the use of anti-dumping** measures to level the playing field and ensure fair trade practices among member nations.
- The WTO does allow member countries to **use anti-dumping duties as a tool to counter unfair pricing by foreign exporters**. These measures are considered **legally valid as long as there is evidence of dumping** causing material injury to the domestic industry and proper investigations are conducted. **So, statement 1 is correct.**
- While anti-dumping duties **aim to protect domestic industries,** they also result in **increased costs for importers**, which are often **passed on to consumers**. This leads to a rise in the price of affected goods in the domestic market. **So, statement 2 is correct.**
- In India, the anti-dumping process is overseen by the **Directorate General of Trade Remedies (DGTR), which is under the Ministry of Commerce and Industry**. However, the **levy of duties is the prerogative of the Ministry of Finance, based on recommendations from DGTR. So, statement 3 is not correct.**
- The amount of anti-dumping duty is **calculated based on the margin of dumping**, i.e., the difference between the export price and the normal value of the good. This is different from a **countervailing duty (CVD),** which is **based on the subsidy provided** by the exporting country. **So, statement 4 is not correct.**

Therefore, option (a) is the correct answer.
Relevance: India recently has imposed anti-dumping duty on five Chinese goods, including vacuum flasks and aluminium foil.

Q83.
Answer: b
Explanation:

- **The Ministry of Micro, Small, and Medium Enterprises (MSME)** revised the classification of MSMEs, **raising both the investment and turnover thresholds**, effective from April 1, 2025. This change is aimed at **expanding the base of MSMEs** by accounting for inflation and the scale of modern enterprises, while still preserving the focus on small-scale industries.
- **For Micro Enterprises**, the investment limit was **revised from ₹1 crore to ₹2.5 crore, and the turnover limit was increased from ₹5 crore to ₹10 crore**. This helps very small firms remain within the benefits of MSME schemes despite expanded operations. **So, row 1 is not correctly matched.**
- **Small Enterprises** are now defined as those with an **investment of up to ₹25 crore (earlier ₹10 crore) and a turnover of up to ₹100 crore (earlier ₹50 crore**). This change significantly broadens the scope for small firms under this category. **So, row 2 is correctly matched.**
- Medium Enterprises are now those with an **investment of up to ₹125 crore (up from ₹50 crore) and a turnover cap of ₹500 crore (up from ₹250 crore),** aligning with the changing structure of industries and their working capital needs. **So, row 3 is correctly matched.**

So, in only two of the above rows is the given information correctly matched.
Therefore, option (b) is the correct answer.

Q84.
Answer: c
Explanation:

- **World Wide Fund for Nature (WWF)** is one of the world's leading conservation organizations, known for pioneering global campaigns and data-driven conservation efforts. Its projects aim to raise awareness, track biodiversity trends, and address environmental degradation. WWF was founded in 1961 and operates in over 100 countries.
- **Earth Hour** is the **world's largest grassroots environmental movement**, initiated by WWF in **2007**. It involves switching off non-essential lights for one hour annually to promote awareness **around energy consumption and climate change**. The event unites people across the world to take symbolic action for the planet. **Earth Hour 2025 focused on water conservation and coincided with World Water Day. So, point 1 is correct.**
- **TX2 Goal** refers to WWF's ambitious conservation target to **double the global population of wild tigers by the year 2022,** from the baseline set in 2010. It involves transboundary cooperation between tiger-range countries, enhanced anti-poaching efforts, and habitat conservation. **So, point 2 is correct.**
- **TRAFFIC** is a joint program of **WWF and** International Union for Conservation of Nature **(IUCN)** that monitors and investigates wildlife trade. It focuses on reducing **illegal wildlife trafficking and promoting sustainable trade practices.** It plays a critical role in wildlife crime prevention globally. **So, point 3 is correct.**
- **Earth Overshoot Day** marks the **date each year when humanity's resource consumption exceeds Earth's capacity to regenerate those resources in that year.** It's a symbolic date calculated by the **Global Footprint Network. So, point 4 is not correct.**

So, only three of the above initiatives are launched by the WWF.
Therefore, option (c) is the correct answer.

Q85.
Answer: a
Explanation:

- The new Standard Operating Procedure (SOP) issued by the **Ministry of Rural Developmen**t lays out clear guidelines to ensure **fair and participatory deletion of job cards** under Mahatma Gandhi National Rural Employment Guarantee Act (**MGNREGA), 2005**. It mandates that every worker flagged for deletion must be given a fair hearing and due process.
- The SOP specifies that all **deletions of job cards must be carried out only after due verification in a Gram Sabha.** This includes a regular Gram Sabha, a social audit Gram Sabha, or a special Gram Sabha convened specifically for reviewing deletions. This ensures community participation and transparency. **So, statement 1 is correct.**
- Programme Officers are required to **display the list of workers proposed for deletion at least 30 days in advance of the Gram Sabha.** The list must be published in public spaces like Panchayat offices, post offices, and ration shops, and also shared directly with workers via SMS or voice calls. This ensures that affected workers have ample time to raise objections.
- If a worker's job card is deleted, they **have the right to appeal the decision within 90 days by submitting a petition to the Block Development Officer (BDO).** The appeal process must be completed within 30 days, providing an institutional safeguard against wrongful deletions. **So, statement 2 is correct.**
- The SOP clearly states that **no deletion of a job card can proceed until all outstanding liabilities — such as unpaid wages, unemployment allowances, or compensation for delays — are fully settled.** This ensures that workers are not denied their rightful dues. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.

Q86.
Answer: c
Explanation:

- **IOS SAGAR (Indian Ocean Ship SAGAR)** is India's latest **maritime initiative in the Southwest Indian Ocean Region (IOR),** operational from April to May, 2025. Led by the Indian **Navy's INS Sunayna, t**he program involves joint activities with nine African nations including Seychelles, Mauritius, and Mozambique. The main objectives are to enhance maritime surveillance, conduct joint EEZ patrols, and strengthen regional security through training and naval cooperation.
- This initiative is **aligned with India's broader SAGAR (Security and Growth for All in the Region) vision, introduced in 2015,** which seeks a safe, inclusive, and prosperous maritime neighborhood. Through joint exercises, port calls, and naval training, IOS SAGAR reinforces India's role as a security provider in the region. It also counters external influences and enhances regional capacity-building through practical engagements.
- The co-hosted exercise **"AIKEYME" (Africa-India Key Maritime Engagement), held in mid-April 2025 in Tanzania,** further supports this effort with table-top simulations, SAR drills, VBSS operations, and maritime law enforcement collaboration. These activities collectively serve to improve interoperability, regional trust, and shared maritime governance.

Therefore, option (c) is the correct answer.

Q87.
Answer: d
Explanation:

- **DNA polymorphisms are variations in DNA that occur in at least 1% of the population.** Common types include Restriction Fragment Length Polymorphisms (**RFLPs),** Variable Number Tandem Repeats **(VNTRs), STRs (microsatellites), and** Single Nucleotide Polymorphisms **(SNPs).** These variations are fundamental in clinical diagnostics, population studies, and criminal investigations. SNPs are now widely used in Genome-Wide Association Studies (GWAS) to identify genes related to complex diseases. DNA sequencing has made the identification of these polymorphisms more rapid and precise.
- **DNA polymorphisms**, specifically SNPs, play a **crucial role in stem cell therapy** by influencing transplant outcomes, **donor selection, and personalized medicine approaches**. They can indicate susceptibility to graft-versus-host disease (GVHD) and other complications. **So, point 1 is correct.**
- **Forensic DNA typing** heavily relies on DNA polymorphisms like **STRs, VNTRs, and SNPs.** These markers provide unique DNA patterns for each individual, which are used to identify suspects or victims with high accuracy in criminal and legal investigations. **So, point 2 is correct.**
- In medical genetics**, disease gene mapping employs polymorphic markers** to track the inheritance of genetic diseases. This allows researchers to locate specific genes associated with inherited conditions across generations within families. **So, point 3 is correct.**
- **Pharmacogenomics** applies DNA polymorphism, particularly SNP analysis, to understand how genetic variations influence an individual's reaction to drugs. This enables more effective and personalized drug prescriptions.
- Polymorphic DNA markers are used in **evolutionary biology to study variation** within and between species. They help in constructing phylogenetic trees and understanding species divergence, migration, and adaptation.
- **Paternity and kinship testing** rely on **comparing polymorphic markers** (like VNTRs and microsatellites) between individuals. The high variability in these markers allows accurate determination of biological relationships. **So, point 4 is correct.**

So, all four of the above statements are correct.
Therefore, option (d) is the correct answer.

Q88.
Answer: a
Explanation:

- **A Geographical Indication (GI) tag** is a unique symbol that identifies products from a specific region or location. It's used to certify that a product has certain qualities, reputation, or characteristics due to its place of origin.
- Also called the **"fireball chilli," Dalle Khursan**i is grown in the **high-altitude regions of Sikkim.** It is intensely pungent and rich in nutrients. GI-tagged in 2020, it has recently gained international attention due to its export to the Solomon Islands. **So, pair 1 is not correctly matched.**
- Grown in the **Khola region of South Goa, Khola Chilli** is noted for its bright red color and distinct aroma. It is a traditional product cultivated by tribal communities and has GI protection for its regional authenticity. **So, pair 2 is not correctly matched.**

- **Bhiwapur Chilli** is cultivated in the **Bhiwapur region of Maharashtra** and is known for its vibrant red color and spiciness. It is widely used in local spice mixes and has high demand across central India. **So, pair 3 is correctly matched.**

So, only one of the pairs given above is correctly matched.
Therefore, option (a) is the correct answer.

Knowledge Box
• **Geographical Indication (GI) tagging** boosts product recognition and helps in preserving regional crop heritage. • The first GI tag was given to **Darjeeling tea in 2004-05.** • **Agricultural and Processed Food Products Export Development Authority (APEDA) facilitates agri-exports** of GI-tagged produce, enhancing farmer incomes and global visibility. • GI recognition leads to premium pricing, direct export access, and better livelihood opportunities for local farmers.

Q89.
Answer: d
Explanation:

- The **Pradhan Mantri Virasat Ka Samvardhan (PM VIKAS)** is a **Central Sector Scheme** of the Ministry of Minority Affairs which **converges five erstwhile schemes** viz. 'Seekho Aur Kamao', 'Nai Manzil', 'Nai Roshni' and 'USTTAD' & 'Hamari Dharohar' schemes and focuses on **upliftment of six notified minority communities. So, statement 1 is correct.**
- It focuses through the following:
  - Skilling and Training (Non-traditional and traditional)
  - Women Leadership and Entrepreneurship
  - Education (through National Institute of Open Schooling)
  - Infrastructure Development (through Pradhan Mantri Jan Vikas Karyakram)
- It mandates that implementing partners ensure the placement of **at least 75% of the total candidates** who are trained under **National Skills Qualifications Framework (NSQF)** aligned skill programs. This provision ensures not just skill training but also tangible employment outcomes for minority communities. **So, statement 3 is correct.**
- The scheme also includes **credit linkage support for beneficiaries** by connecting them to loan programs through the National Minorities Development & Finance Corporation (NMDFC). This aspect is crucial for promoting self-employment and entrepreneurship among skilled individuals, especially artisans and small business owners. **So, statement 2 is correct.**

Therefore, option (d) is the answer.

Q90.
Answer: d
Explanation:

- **Cascade Glacier – United States:** Located in Washington State, this is a glacier on Mount Shuksan in the North Cascades. It is a popular destination for mountaineers and is part of the Cascade Range. It is part of the glacial system in the Pacific Northwest of the U.S. **So, pair 1 is not correctly matched.**

- **Mount Lewotobi – Indonesia:** This is a prominent **stratovolcano located on the island of Flores in Indonesia**. It has twin peaks known as Lewotobi Lakilaki (male) and Lewotobi Perempuan (female). It is one of many volcanoes forming part of the **Pacific Ring of Fire. So, pair 2 is not correctly matched.**
- **Katchatheevu Island – Sri Lanka:** Katchatheevu is a small uninhabited island in the Palk Strait. Although historically accessed by Indian fishermen, it was officially ceded to Sri Lanka through bilateral agreements in the 1970s. **So, pair 3 is not correctly matched.**

So, none of the pairs given above are correctly matched.
Therefore, option (d) is the correct answer.

Q91.
Answer: a
Explanation:

- Inner Line Permit and Administrative District Councils are the special provisions granted to the North Eastern States to help preserve the identity, culture, and governance rights of tribal populations.
- **The Inner Line Permit (ILP)** is an official travel document required by Indian citizens to enter certain protected tribal areas, aimed at safeguarding indigenous communities from external influence.
- **The Autonomous District Councils (ADCs)** are self-governing bodies established under the Sixth Schedule of the Constitution to provide administrative autonomy in tribal areas.
- **Inner Line Permit:**
  - The **Inner Line Permit (ILP)** originates from the **Bengal Eastern Frontier Regulation (BEFR), 1873**, enacted by the **British colonial government**.
  - Introduced to **restrict the entry of British subjects** (including Indians) into certain tribal-dominated areas in the Northeast.
  - **It is aimed to protect commercial interests** like tea, oil, and timber; **isolate tribal populations** from external influence to avoid unrest and maintain **administrative control** over sensitive frontier regions.
  - **Under the 1873 Act,** the ILP was applicable only to three states in the Northeast – **Mizoram, Arunachal Pradesh, and Nagaland.**
  - But in the year 2019, Manipur became the fourth state where the ILP system is implemented.
- **Autonomous District Councils:**
  - The Sixth Schedule of the Constitution deals with the **administration of tribal areas** in the Northeastern states of **Assam, Meghalaya, Tripura, and Mizoram**.
  - **Article 244(2)** extends the provisions of the Sixth Schedule to **the administration of tribal areas** in the states of Assam, Meghalaya, Tripura, and Mizoram according to which the **tribal areas in these 4 states** will be governed under a **special system** — the **Autonomous District Councils (ADCs)** — instead of the usual system followed by the rest of India.
  - **Article 275(1)** (deals with financial assistance to the Tribal Areas) enables parliament to provide **grants-in-aid** from the Consolidated Fund of India for **Strengthening the Autonomous District Councils (ADCs).**

Therefore, option (a) is the correct answer.
Relevance: Indian railways likely to shelve projects in Meghalaya due to local opposition.

Q92.
Answer: c
Explanation:

- The **Euclid mission** is a space telescope developed by the **European Space Agency (ESA)**
- Launched on **July 1, 2023**, Euclid aims to explore the mysterious components of the universe: **dark matter** and **dark energy**, which together constitute about 95% of the universe's total mass-energy content.
- It is expected to **observe over 1.5 billion galaxies**, providing a comprehensive view of the universe's large-scale structure.
- **Mission Objectives:**
  - **Map the Universe in 3D**: Euclid will create a three-dimensional map of the universe by observing billions of galaxies up to 10 billion light-years away.
  - **Understand Cosmic Acceleration**: By studying the distribution and shapes of galaxies, Euclid will help scientists understand **how dark energy influences the universe's accelerated expansion.**
  - **Investigate Dark Matter**: Through **gravitational lensing,** Euclid will measure how **dark matter bends light from distant galaxies**, providing insights into its distribution and properties.
- As of now, Euclid has begun its survey operations and has already captured images containing millions of galaxies. These early observations demonstrate the telescope's capability to achieve its scientific goals.
- Kepler, TESS, or James Webb **involved in the search for habitable exoplanets**. James Webb is also involved in the study of the chemical composition of planetary atmospheres.
- Solar missions like SOHO, Parker Solar Probe, or Aditya-L1 do are **aimed at monitoring solar activity and its impact on Earth's magnetosphere.**

Therefore, option (c) is the correct answer.
Relevance: The first data generated by the European Space Agency's Euclid telescope offers an opportunity to understand the finest structures and distances of lakhs of galaxies.

Q93.
Answer: c
Explanation:

- The **in-house mechanism** was created as a result of the Supreme Court's judgement in ***C.Ravichandran Iyer v Justice A.M.Bhattacharjee (1995)***. The decision noted that it was a mechanism to **fill in the "yawning gap" between proved misbehaviour and bad conduct** inconsistent with the high office. **So, statement 1 is correct.**
- It is used for **allegations of misconduct** against **High Court and Supreme Court judges to ensure judicial accountability**. This process is **completely separate** from the **constitutional procedure** under Article 124(4) of the Constitution, which involves Parliament.
- **In-house Enquiry Mechanism:**
  - Complaints can be submitted to the **Chief Justice of India (CJI)**, a **High Court Chief Justice**, or the **President of India.**
  - If received by a High Court Chief Justice or the President, the complaint is forwarded to the CJI.
  - The CJI evaluates the complaint's seriousness. For credible complaints, the CJI may request a preliminary report from the concerned High Court's Chief Justice.
  - If the preliminary report suggests a deeper investigation, the CJI constitutes a **three-member committee (two Chief Justices from other High Courts, one High Court judge**). For allegations against a Supreme Court judge, the committee comprises three Supreme Court judges. **So, statement 2 is not correct.**

- In conducting the inquiry, the i**n-house committee is free to devise its own procedure** consistent with the **principles of natural justice**. **So, statement 3 is correct.**
- The concerned judge is entitled to appear before this committee and have their say. However, as this is **not a formal judicial inquiry**, there is **no provision for witness examination, cross-examination and legal representation**. This is one of the main reasons the in-house procedure is different from the ones followed by the inquiry committees established by the Speaker of the Lok Sabha or Chairman of the Rajya Sabha under the Judges (Inquiry) Act, 1968. **So, statement 4 is correct.**
- If misconduct is not proven or is minor, the judge is informed, and no further action is taken. If serious misconduct is established, the judge may be asked to resign or seek voluntary retirement.
- If the judge refuses, the matter is escalated to the President and Prime Minister, potentially leading to impeachment proceedings in Parliament.

So, only three of the above statements are correct.
Therefore, option (c) is the correct answer.
Relevance: Chief Justice of India (CJI) Sanjiv Khanna has initiated an unprecedented three-member in-house inquiry into a Delhi High Court judge.

Q94.
Answer: b
Explanation:

- The **National Payments Corporation of India (NPCI) has launched BHIM 3.0,** an upgraded version of the **Bharat Interface for Money (BHIM) app**, enhancing its features to promote digital transactions and financial inclusion across India.
- BHIM is a UPI-based mobile payment application developed by NPCI, facilitating instant bank-to-bank money transfers using mobile numbers or UPI IDs.
- **New Features of BHIM 3.0:**

Feature	Description
Donation Integration	Donate to verified NGOs & institutions directly through app
UPI Mandate Management	Set auto-debit instructions for recurring payments (subscriptions, EMIs, bills)
Enhanced Transaction Limit	Higher UPI transfer limits beyond ₹1 lakh (for select services like IPOs, mandates)
Revamped User Interface	Cleaner layout, smoother navigation for better user experience
Multilingual Support	Available in multiple Indian languages – promotes regional digital inclusion
Low Internet Optimization	Works even on 2G/3G or weak network: USSD (*99#), retry queue, lightweight design
Offline QR Code Support	Merchants can use static QR codes for payments even without internet
New Spending Dashboard	Monthly transaction summary & visual breakdown to track user expenses
Improved Security	Enhanced backend for fraud detection, faster and more secure payments

- **BHIM 3.0** is designed to operate **efficiently in low or no internet conditions**. It achieves this by offering **USSD-based services** alongside its mobile app functionality. This enables users to make **UPI transactions via the *99# code**, without needing an internet connection. **So, statement 1 is correct.**
    - **USSD (Unstructured Supplementary Service Data)** is a protocol that enables **mobile phones to communicate with their service provider's** computers, allowing for real-time interaction and service access.
- The dashboard in BHIM 3.0 provides users with a visual summary of their monthly spending patterns, based exclusively on transactions made through the BHIM App. It **tracks only the spends done via BHIM**, not through other UPI apps or platforms. **So, statement 2 is correct.**
- BHIM 3.0 does not offer a digital wallet feature like Paytm or PhonePe. It is a UPI-based payment application, meaning all transactions are done directly from the user's linked bank account — not through a preloaded wallet. **So, statement 3 is not correct.**

So, only two of the above statements are correct.
Therefore, option (b) is the correct answer.
Relevance: NPCI BHIM Services Limited (NBSL), a wholly owned subsidiary of the NPCI, has launched BHIM 3.0, marking the third evolution of the popular digital payments platform.

Q95.
Answer: d
Explanation:

- **Signal** is a **free, open-source, encrypted messaging app** developed by the **Signal Foundation**. It allows users to send **private messages**, make **secure voice/video calls**, and share media — all protected with **end-to-end encryption**.
- While Signal is more secure than conventional texting, it is **not entirely immune to hacking**.
    - If a user's **mobile device is infected with spyware or malware**, an attacker can potentially read messages even before or after they are encrypted.
    - **Social engineering attacks**, such as phishing, can deceive users into revealing personal information or installing malicious apps.
    - **SIM-swapping attacks** can allow an attacker to hijack a phone number and gain temporary access to verification codes or linked accounts.
    - If someone gains **physical access to an unlocked device**, they can read messages and access Signal despite the encryption. **So, statement 1 is not correct.**
- Signal uses the **Signal Protocol**, one of the most respected and widely adopted **end-to-end encryption (E2EE)** standards. This protocol ensures that the content of messages, voice calls, video calls, and even media shared between users is **encrypted on the sender's device and decrypted only on the recipient's device**. **So, statement 2 is correct.**

So, statement–I is incorrect, but Statement–II is correct.
Therefore, option (d) is the correct answer.
Relevance: Top Trump administration officials used the messaging app Signal to share war plans.

Q96.
Answer: a
Explanation:

- The **Gold Monetisation Scheme (GMS)** was launched by the Government of India in **2015** under the aegis of the Ministry of Finance and implemented by the Reserve Bank of India (RBI).
- The scheme's primary **objective is to mobilize idle gold held by individuals, institutions, and temples, and integrate it into the formal economy. So, statement 1 is correct.**

- Once gold is deposited through the GMS, it is managed by **banks or the Reserve Bank of India (RBI)**. The gold can be used by banks for **loans**, or in other financial systems, which leads to a **multiplier effect** in the economy through the following means.
    - Banks and financial institutions can use the **gold deposits** as collateral for **lending** and other financial activities.
    - It also encourages savings in gold without the need for individuals to physically store or protect their gold.
    - This increases the overall **liquidity** in the economy, as the gold is now part of the **formal financial sector** rather than being locked away in homes or vaults.

Key Features of the GMS includes:

- **Gold Deposit Accounts**: Gold can be deposited in **Short-Term (1 to 3 years) (renewable up to 5 years)**, **Medium-Term (5 to 7 years)**, or **Long-Term (12 to 15 years)** deposit accounts.
    - Effective **March 26, 2025**, the government decided to discontinue the Medium Term and Long Term Government Deposit **components of the GMS.**
    - This decision was attributed to evolving market conditions and the scheme's performance.
    - Consequently, **gold deposits under these categories will no longer be accepted** at designated collection and purity testing centers, GMS mobilization agents, or bank branches. **So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Relevance: The government has recently decided to discontinue the Gold Monetisation Scheme (GMS) in view of the evolving market conditions.

Q97.

Answer: a

Explanation:

- The **DX-EDGE** (Empowering Excellence and Growth through Digital Transformation) initiative is a national program launched by the **NITI Aayog**, supported by the **Ministry of MSME**, to foster collaboration among the **public**, **private**, and **academic sectors**.
- Its primary aim is to create an ecosystem that connects **Medium, Small, and Micro Enterprises (MSMEs)** with **technology providers** and **academic institutions**, thereby facilitating **digital transformation and enhancing global competitiveness**.

Therefore, option (a) is the correct answer.

Knowledge Box
• The model is designed to assist approximately six crore existing MSMEs in India, addressing their challenges through technology and Artificial Intelligence solutions. **Key Objectives of DX-EDGE includes:** • **Empowerment of MSMEs:** DX-EDGE aims to equip MSMEs with the necessary tools, knowledge, and ecosystem to become future-ready and competitive. • **Collaboration Enhancement:** The initiative fosters collaboration among public, private, and academic sectors, facilitating the development of technical expertise and ensuring MSMEs have structured, cost-effective access to technology. • **Integration of Advanced Technologies:** By incorporating technologies such as Artificial Intelligence (AI), DX-EDGE seeks to enhance the global competitiveness of MSMEs, providing innovative solutions to their challenges.

- **Support for Existing MSMEs:** The program extends its benefits to the approximately six crore existing MSMEs in India, addressing their specific challenges through tailored technological solutions.
- **Key Stakeholders:**
  - **Confederation of Indian Industry (CII):** Spearheads the initiative, providing industry insights and support.
  - **NITI Frontier Tech Hub (NITI FTH):** Promotes emerging technologies and innovation.
  - **All India Council for Technical Education (AICTE):** Engages academic institutions in facilitating digital education and training.
- **DX-EDGE** plays a crucial role in driving inclusive digital adoption and accelerating the vision of Viksit Bharat 2047.

Relevance: NITI Aayog CEO BVR Subrahmanyam recently launched DX-EDGE-Empowering excellence and growth through digital transformation.

Q98.
Answer: b
Explanation:

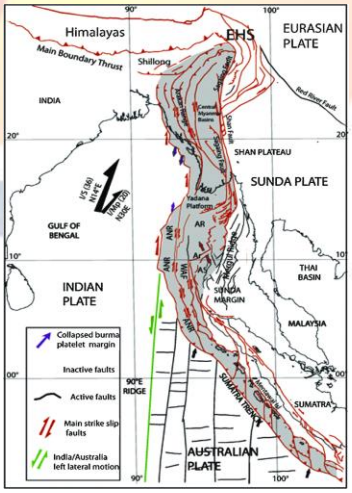
- **Folk dances are an essential part of India's intangible cultural heritage,** reflecting the lifestyle, traditions, beliefs, and local customs of various communities across the country.
- They are generally performed **during festivals, harvests, religious rituals, and social celebrations,** often involving local instruments, traditional costumes, and group participation.
- **Theyyam - Kerala:**
  - Theyyam is believed to have originated over **800 years ago**, representing a form of Hinduism practiced by tribal communities.
  - Includes **ritual dance, music, and recitation of mythological stories**.
  - Theyyam serves as a medium for **ancestor worship** and **hero worship**, reflecting the beliefs of ancient tribal communities.
  - Performers (mostly from lower castes) don elaborate **makeup, face painting, headgear**, and costumes.
- **Fugdi - Goa:**
  - Fugdi is a folk dance performed **exclusively by women in Goa** and parts of Konkan, especially during Ganesh Chaturthi and Vrata rituals.
  - It reflects the **agrarian and ritualistic life of women, particularly from the Dhangar (shepherd) community.**
  - Dancers form circles or rows and maintain rhythm through clapping and blowing **air ("foo" sound), without musical instruments.**
  - Variants like Kalashi Fugdi (with pitchers) and Katti Fugdi (with coconut shells) highlight daily life and devotional themes.
- **Thougal Jagoi - Manipur:**
  - A ritual dance form of the **Meitei community in Manipur,** performed during the **Lai Haraoba festiva**l to honor local deities and narrate creation myths.
  - It is a **devotional offering to the Umang Lai (local deities)** and is integral to Meitei religious cosmology.
  - Performed by **Maibis (priestesses) and male dancers**, typically in slow, symbolic movements.
  - It emphasizes ritual purity and the cosmic role of deities i**n the Meitei belief system.**
  - Reflects **Meitei mythology, ancestral worship**, and the spiritual connection between humans and deities.

- **Dangi - Gujarat:**
  - A tribal dance performed by the **Dangi tribe in the Dang district of Gujarat and parts of Maharashtra**, reflecting the forest-dwelling lifestyle and connection with nature worship.
  - Celebrated during harvest festivals, marriages, and community events, symbolizing gratitude to nature and seasonal cycles.
  - Dancers form circles or semi-circles, using rhythmic footwork and hand movements.
  - Accompanied by traditional tribal instruments like **Dhol, Pavri, and Shehnai.**

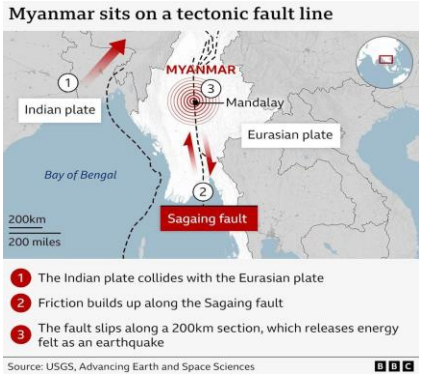
So, only two of the above pairs are correctly matched.
Therefore, option (b) is the correct answer.

Q99.
Answer: d
Explanation:

- Myanmar is situated in one of the **most tectonically complex regions** in the world. The seismic activity here is due to the interaction of several major and micro tectonic plates. The key players in this system are:
- **Indian Plate:**
  - Moves northeastward.
  - Collides with the **Eurasian Plate**, leading to the uplift of the Himalayas and causing immense stress in the region.
  - Also subducts (dives under) beneath the **Burma Plate** along the **Arakan Trench** in western Myanmar.
- **Eurasian Plate**
  - Lies to the north of Myanmar.
  - Interaction with the Indian Plate creates compressional forces.
  - The **recent earthquake in Myanmar occurred along the Sagaing Fault**, a major tectonic boundary **between the Indian Plate and the Eurasian Plate**. T**he Burma Microplate, situated between these two larger plates**, also played a significant role in the region's seismic activity.



- **Burma Plate:**
  - Also known as the **Burma Microplate**.
  - Sandwiched between the **Indian Plate** to the west and the **Sunda Plate** to the east.
  - Home to the **Sagaing Fault**, a major strike-slip fault that runs north-south through central Myanmar.



- **Sunda Plate:** Located southeast of the Burma Plate.While the **Sunda Plate interacts** with the Burma Plate **in Indonesia and nearby areas**. The Sagaing Fault in Myanmar is a major fault and serves as a mainly continental right-lateral transform **fault between the Indian plate and Sunda plate.**

Therefore, option (d) is the correct answer.

Q100.
Answer: b
Explanation:

- **The International Union for Conservation of Nature (IUCN)** has introduced a new framework called the **Green Status of Species,** which assesses not only the risk of extinction but **also the recovery status of species.**
- The new assessment tool is divided into **nine recovery categories:** Fully recovered, slightly depleted, moderately depleted, largely depleted, critically depleted, extinct in the wild, and indeterminate.
- There are now over **100 IUCN Green Status of Species assessments on the IUCN Red List.**
- The Green Status **complements the IUCN Red List** by providing a **tool for assessing the recovery of species** and measuring their conservation success.

Therefore, option (b) is the correct answer.

Knowledge Box
• In the first Green Status assessment**, lions in Africa and India have been categorized as 'Largely Depleted',** which indicates that despite conservation efforts, its population is still significantly lower than its historical numbers and is restricted to a small area. • The Asiatic lion is **classified as vulnerable on the IUCN Red List**, which means it is at risk of extinction in the wild. • The assessment shows that conservation has prevented likely extinctions from West and Southern Central Africa, South Africa and India. Conservation efforts, **particularly in Africa and India** (such as anti-poaching, habitat preservation, and community engagement), have helped **prevent the extinction of lions**. For example, the Asiatic lion's population has been steadily increasing in the Gir Forest after successful conservation initiatives. • Lions are **not fully ecologically functiona**l across their entire range because their population is fragmented, and their presence in many areas is not enough to restore or maintain ecological balance. For instance, lions' absence from many parts of their historical range and conflicts with human activities have limited their ecological role.

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